

TokTickIT Lab 2

Final Submission Evidence

Student: Chartanat Upthaipiboon | Student ID: 67070507210

Repository: [Chxtamos/-TokTickIT-](#)

Release merge commit: 7780ce098f5e8ae690dfa30e0e962dccc6ba9033

Final delivery documentation commit: 6ad3605af01c60dc2be2e006608c15df2d6f8eb6

This PDF follows the Lab 2 labsheet exactly: one PDF, working links, readable screenshot evidence, and headings Answer Part 1 through Answer Part 9 in the required order.

Answer Part 1: Git Use with Engineering Workflow

Lab 2 used the required staged workflow: completed Lab 1 main -> lab2-staging -> one feature branch per Issue -> peer-reviewed Pull Request -> lab2-staging -> one release Pull Request -> main.

Repository: [GitHub repository](#)

Final release: [PR #50 - Release: TokTickIT Lab 2](#)

Peer-review record: [reviewer.md](#)

Project board: [TokTickIT Individual Sprints](#)

Final feature PRs #13, #15, #17, #19, #21, #23, #25, #27, #29, #31, #33, #35, #37, and #44-#49 target lab2-staging. All Issues are closed after review; PR #50 merged lab2-staging into main.

Repository structure and ignored-file evidence:

toktickit/ client/src, client/tests/lab-01, client/tests/lab-02, client/e2e/lab-02 server/src, server/tests/lab-01, server/tests/lab-02, server/prisma docs/lab-02, artifacts/lab-02/screenshots, output/pdf

The repository .gitignore excludes node_modules, .env files, build/dist output, Playwright reports/results, and e2e temporary uploads. README and .gitignore are rendered and linked in the final repository: [README.md](#) | [.gitignore](#)

Final main commit and merge history

317a871 docs: align Lab 2 PDF with official labsheet
6ad3605 docs: complete Lab 2 final delivery
7780ce0 Merge pull request #50 from Chxtamos/lab2-staging
ae147e8 Merge pull request #49 from Chxtamos/feature/23-lab2-final-verification-release
6b67a23 Merge pull request #48 from Chxtamos/feature/22-lab2-visual-evidence
dce9816 Merge pull request #47 from Chxtamos/feature/21-lab2-responsive-visual
453d14e Merge pull request #46 from Chxtamos/feature/20-lab2-e2e-attachment-lifecycle
3bccfb0 Merge pull request #45 from Chxtamos/feature/19-lab2-e2e-my-tickets-detail
cda7b34 Merge pull request #44 from Chxtamos/feature/18-lab2-e2e-requester-ticket
...
44c96a9 Merge pull request #13 from Chxtamos/feature/5-lab2-engineering-contract

All implementation Issues and the final verification Issue are closed. The linked GitHub Project provides the final Kanban evidence with completed items.

Answer Part 1 - GitHub Screenshot Evidence

Final main commit history - section 1 of 2

PlatformSolutionsResourcesOpen SourceEnterprisePricing

Sign inSign up

Chxtamos / -TokTickIT- Public

NotificationsFork 0Star 0

<> CodeIssuesPull requestsActionsProjectsSecurity and qualityInsights

main

All usersAll time

Commits on Sep 5, 2026

docs: complete Lab 2 final delivery
Chxtamos committed 10 hours ago

6ad3605

<>

Merge pull request #50 from Chxtamos/lab2-staging

Tanaboonnnnn authored 10 hours ago

7780ce0

<>

Merge pull request #49 from Chxtamos/feature/23-lab2-final-verification-release

Tanaboonnnnn authored 11 hours ago

ae147e8

<>

chore: record Lab 2 final verification
Chxtamos committed 12 hours ago

9c2ae2e

<>

Merge pull request #48 from Chxtamos/feature/22-lab2-visual-evidence

Tanaboonnnnn authored 12 hours ago

6b67a23

<>

feat: add Lab 2 visual screenshot evidence
Chxtamos committed 14 hours ago

ae0c8b8

<>

Merge pull request #47 from Chxtamos/feature/21-lab2-responsive-visual

Tanaboonnnnn authored 14 hours ago

dce9816

<>

test: strengthen responsive evidence
Chxtamos committed 15 hours ago

cba5b98

<>

Commits on Sep 4, 2026

feat: add responsive viewport E2E coverage
Chxtamos committed yesterday

1c58d15

<>

Merge pull request #46 from Chxtamos/feature/20-lab2-e2e-attachment-lifecycle

Tanaboonnnnn authored yesterday

453d14e

<>

Commits on Sep 3, 2026

test: verify active attachment ownership protection
Chxtamos committed 3 days ago

91234d8

<>

feat: add attachment lifecycle E2E coverage
Chxtamos committed 3 days ago

e44e44c

<>

Merge pull request #45 from Chxtamos/feature/19-lab2-e2e-my-tickets-detail

Tanaboonnnnn authored 3 days ago

3bcfb0

<>

test: strengthen My Tickets E2E evidence
Chxtamos committed 3 days ago

df6fc9c

<>

feat: add My Tickets ownership E2E coverage
Chxtamos committed 3 days ago

da68eaf

<>

Merge pull request #44 from Chxtamos/feature/18-lab2-e2e-requester-ticket

Tanaboonnnnn authored 3 days ago

cda7b34

<>

fix: align feature 18 scope and e2e evidence
Chxtamos committed 3 days ago

f1fac79

<>

Answer Part 1 - GitHub Screenshot Evidence

Final main commit history - section 2 of 2

feat: add requester ticket creation e2e coverage

Commits on Sep 2, 2026

Merge pull request #37 from Chxtamos/feature/17-lab2-style-accessibility

fix: align accessibility evidence with test coverage

feat: add Zen Green accessibility evidence

Merge pull request #35 from Chxtamos/feature/16-lab2-attachment-ui

fix: close attachment UI review gaps

feat: add attachment lifecycle UI

Merge pull request #33 from Chxtamos/feature/15-lab2-ticket-detail

fix: align ticket detail feedback coverage

Commits on Sep 1, 2026

feat: add requester ticket detail screen

Merge pull request #31 from Chxtamos/feature/14-lab2-my-tickets

fix: align my tickets responsive evidence

feat: add my tickets screen

Merge pull request #29 from Chxtamos/feature/13-lab2-create-ticket-ui

Commits on Aug 31, 2026

fix: reset request id for new tickets

fix: make create ticket retries idempotent

feat: add create ticket screen

Merge pull request #27 from Chxtamos/feature/12-lab2-requester-selection

< Previous

Next >

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Answer Part 1 - GitHub Screenshot Evidence

Closed Lab 2 Issues - final Done evidence

The screenshot shows the GitHub interface for the repository **Chxtamos / -TokTickIT**. The repository is public and has 0 forks and 0 stars. The **Issues** tab is selected, showing a list of 12 closed issues. The issues are sorted by state (Open, Closed) and are all marked as closed. The issues are:

- Complete Lab 2 final verification and release (#43) - closed on Sep 5, 2026
- Create Lab 2 screenshot evidence and visual checklist (#42) - closed on Sep 5, 2026
- Add Lab 2 responsive and visual viewport tests (#41) - closed on Sep 5, 2026
- E2E Attachment lifecycle flow (#40) - closed on Sep 4, 2026
- E2E My Tickets ownership and Ticket Detail flow (#39) - closed on Sep 3, 2026
- E2E requester and ticket creation flow (#38) - closed on Sep 3, 2026
- Implement Lab 2 Zen Green style and accessibility evidence (#36) - closed on Sep 2, 2026
- Implement Lab 2 Attachment lifecycle UI (#34) - closed on Sep 2, 2026
- Implement Lab 2 Requester Ticket Detail screen (#32) - closed on Sep 2, 2026
- Implement Lab 2 My Tickets screen (#30) - closed on Sep 2, 2026
- Implement Lab 2 Create Ticket UI (#28) - closed on Sep 1, 2026
- Lab 2: Implement Development Requester selection and application shell (#26) - closed on Aug 31, 2026

The footer of the page shows the GitHub logo, copyright information (© 2026 GitHub, Inc.), and links to Terms, Privacy, Security, Status, Community, Docs, Contact, Manage cookies, and Do not share my personal information.

Answer Part 1 - GitHub Screenshot Evidence

Release PR #50 and peer approval - section 1 of 3

Platform Solutions Resources Open Source Enterprise Pricing

Chxtamos / -TokTickIT- Public

Notifications Fork 0 Star 0

<> Code Issues Pull requests Actions Projects Security and quality Insights

Release: TokTickIT Lab 2 #50

Merged Tanaboornnnnn merged 58 commits into main from lab2-staging 10 hours ago

Conversation Commits 58 Checks Files changed

Chxtamos commented 11 hours ago

Summary

เตรียม Release TokTickIT Lab 2 จาก lab2-staging เข้า main

Verification

- Server build และ tests ผ่าน
- PostgreSQL integration tests ผ่าน
- Prisma migration ผ่าน
- Seed ทำงานเข้าได้โดยไม่มีข้อมูลซ้ำ
- Client tests และ build ผ่าน
- Full Playwright E2E ผ่าน 9 tests
- E2E-01 ถึง E2E-06 ผ่าน
- Visual checklist และ screenshot evidence ครบ
- ไม่พบ secrets, generated output หรือ uploaded files

Release Flow

lab2-staging → main

Chxtamos and others added 30 commits 2 weeks ago

docs: define Lab 2 engineering contract 141a10f

docs(lab2): define engineering contract and test plan 7684a7a

Merge pull request #13 from Chxtamos/feature/5-lab2-engineering-contract 44c96a9 Verified

feat: add Lab 2 data model and seed data 4693730

test: enforce normalized requester email and seed data 4f3fa5e

Merge pull request #15 from Chxtamos/feature/6-lab2-data-model-and-seed 6d4d4ba Verified

feat: add Lab 2 reference data APIs 8147b3c

test: cover reference data exclusions and failures 7fd74c8

Merge pull request #17 from Chxtamos/feature/7-lab2-requester-context... 33bfe18 Verified

Reviewers

Tanaboornnnnn ✓

cottonlnwza

L0u1ss

Peepipat-Suesongnuen

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

3 participants

TokTickIT Lab 2 - 67070507210

Git evidence 4/6

Answer Part 1 - GitHub Screenshot Evidence

Release PR #50 and peer approval - section 2 of 3

feat: add Lab 2 ticket creation API

test: cover concurrent ticket idempotency replay

fix: close Lab 2 ticket API review feedback

Merge pull request #19 from Chxtamos/feature/8-lab2-ticket-create-api

feat: add My Tickets API

Merge pull request #21 from Chxtamos/feature/9-lab2-my-tickets-api

lab2 : Lab 2: Add Ticket Detail API

fix: validate active requester for ticket details

docs: record ticket detail verification evidence

Merge pull request #23 from Chxtamos/feature/10-lab2-ticket-detail-api

Lab 2: Implement Attachment Lifecycle API

fix: harden attachment upload lifecycle

test: verify attachment storage rollback

Merge pull request #25 from Chxtamos/feature/11-lab2-attachments-api

feat: add requester selection and application shell

fix: align requester selector evidence and navigation

Merge pull request #27 from Chxtamos/feature/12-lab2-requester-selection

feat: add create ticket screen

fix: make create ticket retries idempotent

fix: reset request id for new tickets

Merge pull request #29 from Chxtamos/feature/13-lab2-create-ticket-ui

1d034aa

55adfe3

✓ 8d7496a

Verified ✓ 5af8777

✓ 212507e

Verified ✓ 3c8dc1a

✓ 925a4a1

3e8e6c2

✓ cd6a74c

Verified ✓ 22679b9

✓ 28e2ef6

✓ 59a703e

✓ f8c6d76

Verified ✓ cd48d4e

✓ 843455f

✓ dc11b21

Verified ✓ 22f4673

✓ 5611aa7

✓ 2fc6c9c

✓ 3adf76e

Verified ✓ 6d96c9e

5 hidden items

Load more...

Chxtamos and others added 24 commits 4 days ago

fix: align ticket detail feedback coverage

Merge pull request #33 from Chxtamos/feature/15-lab2-ticket-detail

feat: add attachment lifecycle UI

fix: close attachment UI review gaps

Merge pull request #35 from Chxtamos/feature/16-lab2-attachment-ui

feat: add Zen Green accessibility evidence

fix: align accessibility evidence with test coverage

Merge pull request #37 from Chxtamos/feature/17-lab2-style-accessibility

feat: add requester ticket creation e2e coverage

✓ 0eae31

Verified ✓ 40fc021

✓ ef5a4d8

✓ 106e28a

Verified ✓ 6724cc6

✓ c0fe54c

✓ a72c2d8

Verified ✓ 60388e3

✓ 7506047

Answer Part 1 - GitHub Screenshot Evidence

Release PR #50 and peer approval - section 3 of 3

fix: align feature 18 scope and e2e evidence

Merge pull request #44 from Chxtamos/feature/18-lab2-e2e-requester-ti...

feat: add My Tickets ownership E2E coverage

test: strengthen My Tickets E2E evidence

Merge pull request #45 from Chxtamos/feature/19-lab2-e2e-my-tickets-d...

feat: add attachment lifecycle E2E coverage

test: verify active attachment ownership protection

Merge pull request #46 from Chxtamos/feature/20-lab2-e2e-attachment-1...

feat: add responsive viewport E2E coverage

test: strengthen responsive evidence

Merge pull request #47 from Chxtamos/feature/21-lab2-responsive-visual...

feat: add Lab 2 visual screenshot evidence

Merge pull request #48 from Chxtamos/feature/22-lab2-visual-evidence

chore: record Lab 2 final verification

Merge pull request #49 from Chxtamos/feature/23-lab2-final-verificati...

✓ f1fac79

Verified ✓ cda7b34

✓ da68eaf

✓ df6fc9c

Verified ✓ 3bccfb0

✓ e44e44c

✓ 91234d8

Verified ✓ 453d14e

✓ 1c58d15

✓ cba5b98

Verified ✓ dce9816

✓ ae0c8b8

Verified ✓ 6b67a23

✓ 9c2ae2e

Verified ✓ ae147e8

Chxtamos requested review from L0u1sss, Peepipat-Suesoongnuen, Tanaboonnann and cottonlnwza 11 hours ago

Tanaboonnann approved these changes 10 hours ago

View reviewed changes

Tanaboonnann left a comment

Collaborator ...

ตัวเรา ค่าไอเช็ก Release PR #50 รอบสุดท้ายแล้วนะ 🙏🏻
ตอนนี้ flow ถูกต้องแล้วค่า เป็น lab2-staging → main จังตาม release plan หลังจาก final verification ใน PR #49 เข้า staging เรียบร้อยแล้ว

Exact release head ตอนนี้คือ [ae147e8](#) ... แล้วค่าเช็ก CI บน head นี้ให้แล้ว ผ่านครบทั้ง client / server / e2e ✓

Scope กับ release evidence ที่รวมมาจาก staging ก็ตรงกับที่เตรียมไว้ ทั้ง server/client verification, PostgreSQL integration, migration/seed, full Playwright, E2E-01 ถึง E2E-06 แล้วก็ visual evidence

รอบนี้ค่าไม่เจอ blocker เพิ่มแล้วค่า ตัว release PR พร้อม merge เข้า main ได้หลังได้รับ peer approval แล้ว 🙏🏻
Approved ได้เลย

Tanaboonnann merged commit 7780ce0 into main 10 hours ago

View details

6 checks passed

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Rendered copy: README.md

[README.md](#)

TokTickIT

TokTickIT is a requester-facing IT service desk MVP for CPE 334 Lab 2. A tester selects a seeded Development Requester, creates Tickets, views only owned Tickets, and manages permitted Attachments.

Technology Stack

- Frontend: React, TypeScript, Vite, Bootstrap
- Backend: Node.js, Express, TypeScript
- Database and ORM: PostgreSQL, Prisma
- Testing: Vitest, Supertest, React Testing Library, Playwright

Project Structure

```
toktickit/
├── client/
│   ├── src/
│   ├── tests/
│   │   ├── lab-01/
│   │   ├── lab-02/
│   │   └── e2e/lab-02/
│   ├── .env.example
│   └── package.json
├── server/
│   ├── prisma/
│   │   ├── migrations/
│   │   ├── schema.prisma
│   │   └── seed.ts
│   ├── src/
│   ├── tests/
│   │   ├── lab-01/
│   │   │   ├── health.test.ts
│   │   │   └── categories.test.ts
│   │   ├── .env.example
│   │   └── package.json
├── docs/
│   ├── lab-02/
│   │   ├── ai-use.md
│   │   ├── api-spec.md
│   │   ├── reviewer.md
│   │   ├── specification.md
│   │   ├── tests.md
│   │   └── ui-spec.md
├── artifacts/lab-02/screenshots/
├── output/pdf/67070507210_Lab2_Final.pdf
├── .gitignore
└── README.md
```

Prerequisites

Install the following software before starting:

- Node.js and npm
- PostgreSQL 17
- Git

Clone the Repository

```
git clone https://github.com/Chxtamos/-TokTickIT-.git
cd "-TokTickIT-"
```

Install Dependencies

Install frontend dependencies:

```
cd client
npm install
cd ..
```

Install backend dependencies:

```
cd server
```

```
npm install
cd ..
```

Environment Setup

Copy the provided environment examples (do not commit the resulting `.env` files):

```
Copy-Item client\.env.example client\.env
Copy-Item server\.env.example server\.env
```

Configure `server/.env` with your local PostgreSQL connection:

```
DATABASE_URL="postgresql://USER:PASSWORD@localhost:5432/toktickit?schema=public"
```

The frontend API URL can be configured in `client/.env`:

```
VITE_API_URL=http://localhost:3000
```

Do not commit `.env` files, database passwords, or `node\_modules`.

Database Setup

Create a PostgreSQL database named `toktickit` and provide a database user with access to it. This can be done through pgAdmin or PostgreSQL `psql`.

After configuring `DATABASE\_URL`, enter the server directory:

```
cd server
```

Generate the Prisma Client:

```
npx prisma generate
```

Apply the committed database migrations:

```
npx prisma migrate deploy
```

Seed the Lab 2 reference data (safe to run repeatedly):

```
npm run prisma:seed
npm run prisma:seed
```

Check the migration status:

```
npx prisma migrate status
```

The seed is idempotent and can run repeatedly without creating duplicate categories, systems, or Development Requesters.

For development, create a new migration after changing `schema.prisma`:

```
npx prisma migrate dev --name migration-name
```

Running the Application

Start the backend:

```
cd server
npm run dev
```

The backend runs at:

```
http://localhost:3000
```

Start the frontend in another terminal:

```
cd client
npm run dev
```

Open the Vite URL shown in the terminal, normally:

```
http://localhost:5173
```

Attachment files are written to a private, non-public directory. Set `ATTACHMENT\_STORAGE\_DIR` when needed; otherwise the server uses `server/storage/attachments`. Uploaded files and storage paths must never be committed.

REST API

Health Check

GET /api/health

Expected response:

```
{
  "status": "ok",
  "service": "TokTickIT API"
}
```

Reference Data

GET /api/categories

The application exposes active-only Categories, Related Systems, and Development Requesters endpoints. All Ticket and Attachment endpoints require the temporary `X-Requester-Id` context header; this is a Lab 2 test mechanism, not authentication.

Ticket and Attachment workflows

The REST API supports idempotent Ticket creation, owner-scoped My Tickets and Ticket Detail retrieval, active/removed Attachment metadata, validated upload/download, and soft removal. See `docs/lab-02/api-spec.md` for the normative request/response contract and safe error behavior.

Running Tests

Backend tests (including PostgreSQL integration):

```
$env:RUN_DB_INTEGRATION="1"
npm --prefix server test
```

Client unit/UI tests:

```
npm --prefix client test
```

Playwright E2E (requires PostgreSQL and Chromium):

```
npx playwright install chromium
npm --prefix client run e2e
```

The final Lab 2 evidence currently records 62 Server tests, 54 Client tests, and 9 Playwright E2E tests passed. Full traceability and screenshot evidence are documented in `docs/lab-02/tests.md` and `artifacts/lab-02/screenshots/`.

Lab 2 documentation

- `docs/lab-02/specification.md` - approved engineering contract and Definition of Done.
- `docs/lab-02/api-spec.md` - normative REST API contract.
- `docs/lab-02/ui-spec.md` - UI, accessibility, and responsive contract.
- `docs/lab-02/tests.md` - test plan, final results, traceability, and visual checklist.
- `docs/lab-02/reviewer.md` - peer-review and release record.
- `docs/lab-02/ai-use.md` - selected prompts and student reflection.
- `output/pdf/67070507210\_Lab2\_Final.pdf` - concise final delivery report using Answer Part 1 through Answer Part 9.

Rendered copy: .gitignore

.gitignore

dependencies

node_modules/

env & secrets

.env

*.env

!.env.example

build output

dist/

build/

prisma

server/prisma/*.db

logs & OS

*.log

.DS_Store

playwright-report/

test-results/

e2e/.tmp/

Rendered copy: reviewer.md

<docs/lab-02/reviewer.md>

Lab 2 Peer Review Record

Author

- **Name:** Chartanat Upthaipiboon
- **Student ID:** `67070507210`
- **GitHub:** [@Chxtamos](https://github.com/Chxtamos)(<https://github.com/Chxtamos>)
- **Repository:** [\[Chxtamos/-TokTickIT-\]\(https://github.com/Chxtamos/-TokTickIT-\)](https://github.com/Chxtamos/-TokTickIT-)

Lab 2 Reviewer

- **Name:** Tanboon Teawsawat
- **Student ID:** `67070507211`
- **GitHub:** [@Tanaboannnnn](https://github.com/Tanaboannnnn)(<https://github.com/Tanaboannnnn>)

Engineering Contract Issue

Item	Link / Value	Current state
---	---	---
Issue	[#11 — Lab 2 Engineering Contract and Test Plan](https://github.com/Chxtamos/-TokTickIT-/issues/11)	Closed
Required branch	`feature/5-lab2-engineering-contract`	Pushed
Target branch	`lab2-staging`	Exists
Project	`TokTickIT Individual Sprints`	Added
Project status	Completed	Issue and PR workflow completed

Pull Requests and Review Flow

All Feature PRs #13, #15, #17, #19, #21, #23, #25, #27, #29, #31, #33, #35, #37, and #44-#49 were reviewed and merged into `lab2-staging`. PR #12 was closed without merge and superseded by PR #13.

The review conversations document requested changes and follow-up corrections for API contracts, validation, ownership, Attachment lifecycle, accessibility, E2E evidence, responsive behavior, visual evidence, and final verification.

Review Evidence

- PR #12 was closed without merge after the first contract attempt.
- PR #13 corrected the contract documentation and was merged after review.
- Later review rounds requested concrete corrections before approval; the corresponding commits and updated evidence are visible in PR #15 through PR #49.
- All final Feature PRs target `lab2-staging`, and the release PR #50 targets `main`.

Release Review

[PR #50 — Release: TokTickIT Lab 2](https://github.com/Chxtamos/-TokTickIT-/pull/50) merged `lab2-staging` into `main` after Client CI, Server CI, and E2E CI passed.

- Final `main` commit: `7780ce098f5e8ae690dfa30e0e962dcc6ba9033`
- Final verification evidence: `docs/lab-02/tests.md`
- Screenshot evidence: `artifacts/lab-02/screenshots/`

Human Verification Checklist

- [x] The author and reviewer Student IDs are confirmed by the student.
- [x] The spelling of all names is confirmed by the relevant student/reviewer.
- [x] Every recorded PR link opens the correct Pull Request.

- [x] Review comments and requested corrections are recorded in the linked GitHub conversations.
- [x] Every claimed correction appears in the relevant commit/diff.
- [x] Every approval was provided by a real peer reviewer.
- [x] No AI-generated review is presented as peer-review evidence.

Answer Part 2: Spec-Driven Development

The Engineering Contract was completed before implementation and kept as the source of truth. It contains numbered Functional Requirements, Business Rules, UI rules, data decisions, API contract, Acceptance Criteria, and Definition of Done.

Rendered contract: <docs/lab-02/specification.md>

The contract covers the temporary Development Requester testing context, Ticket creation, My Tickets, Ticket Detail, Attachment lifecycle, ownership, validation, failure states, responsive UI, accessibility, peer review, visual inspection, and final main verification.

Pre-implementation evidence: [contract PR #13](#) was completed and merged before the implementation PR sequence #15 onward.

Key approved decisions include server-generated unique Ticket Numbers, owner context through X-Requester-Id, safe 404 for non-owned resources, idempotent creation, five active Attachment limit, soft removal, and no Lab 3 authentication or IT Staff workflow.

Rendered copy: docs/lab-02/specification.md

[docs/lab-02/specification.md](#)

Lab 2 Sprint Engineering Specification

1. Sprint Goal

Deliver a professional, responsive Requester-facing TokTickIT MVP. A selected Development Requester can create an IT support Ticket, receive a backend-generated official Ticket Number, locate and inspect only their own Tickets, and manage permitted Attachments. The sprint extends the Lab 1 React, Express, Prisma, and PostgreSQL foundation while establishing reusable Zen Green UI and evidence-driven engineering conventions.

2. Stakeholder Request Interpretation

Lab 2 replaces the Lab 1 connectivity demonstration with a complete Requester workflow. Real authentication is deferred to Lab 3, so a tester first selects one seeded active Development Requester. That temporary identity becomes the context for Ticket creation, My Tickets, owned Ticket Detail, and Attachment operations. Completion requires working software plus an approved specification, traceable tests, GitHub Issues, feature branches, peer-reviewed Pull Requests, visual inspection, and final evidence.

3. Scope

Included

- Development Requester Selection and Change Requester flows.
- Active Requester, Category, and Related System data loaded from PostgreSQL.
- Create Ticket with a backend-generated Ticket Number and initial 'NEW' status.
- My Tickets containing only the selected Requester's Tickets.
- Search, filtering, sorting, and pagination.
- Requester-owned, read-only Ticket Detail.
- Attachment upload, metadata, download, and soft removal with a reason.
- Backend ownership checks for direct Ticket and Attachment requests.
- Initial, loading, submitting, success, validation, empty, no-results, and safe failure states.
- Reusable Zen Green UI, responsive layouts, and accessibility behavior.
- Unit, API/integration, UI component, UI style, responsive, visual, and E2E evidence.
- GitHub Issues, feature branches, PR review, staged integration, documentation, and final submission evidence.

Excluded

- Real login/logout, passwords, password hashing, sessions, tokens, authenticated identity, and production role-based authorization.
- IT Staff dashboard/queue, claiming, reassignment, and changing IT Priority.
- Public Comments, Internal Notes, Actions Taken/Service Actions, and Event Log features.
- Ticket status transitions after the initial 'NEW' state, including resolve, close, reopen, or cancel.
- Resolution confirmation and resolution-summary workflows.
- Administrator screens for users, roles, Requesters, Categories, or Related Systems.
- Production deployment, malware scanning, email notifications, and cloud object storage.

The sample Ticket Detail image is a visual-direction reference only. Features visible in the image but explicitly excluded above must not be implemented in Lab 2.

4. Functional Requirements

Development Requester Context

- **FR-01:** When no valid Requester is selected, the application shall show the Development Requester Selection screen before any requester-specific screen.
- **FR-02:** The selector shall load only active Development Requesters from PostgreSQL and provide loading, empty, and safe failure states.
- **FR-03:** Continue shall remain disabled until an active Requester is selected.
- **FR-04:** The selected Requester ID shall be stored in browser 'sessionStorage' for the current tab and revalidated when the application starts.

- **FR-05:** The application shell shall display the selected Requester's name and provide a Change Requester action.
- **FR-06:** Changing Requester shall clear requester-owned client state and reload data in the new context.

Ticket Creation

- **FR-07:** Create Ticket shall display pending/read-only Ticket Number and Ticket Date fields and the selected Requester as read-only.
- **FR-08:** Active Categories and Related Systems shall be loaded from PostgreSQL.
- **FR-09:** A Requester shall enter Category, Related System, Summary, Requested Priority, Description, and zero to five permitted Attachments.
- **FR-10:** Frontend validation shall provide field-level feedback, and backend validation shall independently enforce the approved rules.
- **FR-11:** The backend shall create the Ticket for the selected Requester, generate a unique Ticket Number, set status to 'NEW', and return authoritative saved values.
- **FR-12:** Duplicate Ticket creation shall be prevented using a client-generated request ID as well as a busy/disabled Submit control.
- **FR-13:** A failed create request shall preserve editable form values and selected valid files.
- **FR-14:** After Ticket creation, selected files shall upload individually. Failed file uploads shall not roll back the Ticket and shall be retryable.

My Tickets

- **FR-15:** My Tickets shall return and display only Tickets owned by the selected Requester.
- **FR-16:** Search shall support official Ticket Number and Summary.
- **FR-17:** Filters shall support Category, Related System, Requested Priority, and Current Status.
- **FR-18:** The list shall support documented sorting with deterministic secondary ordering.
- **FR-19:** The list shall support one-based pagination and documented page sizes.
- **FR-20:** My Tickets shall have distinct loading, empty, no-results, invalid-query, and safe failure states.
- **FR-21:** Each listed Ticket shall provide a clear action to open Ticket Detail.

Ticket Detail and Attachments

- **FR-22:** Ticket Detail shall display the owned Ticket as read-only information.
- **FR-23:** A direct request for a missing or non-owned Ticket shall not return Ticket data.
- **FR-24:** Ticket Detail shall list active and removed Attachment metadata and distinguish their states.
- **FR-25:** The owner shall be able to upload a permitted Attachment while fewer than five active Attachments exist.
- **FR-26:** The owner shall be able to download an active Attachment.
- **FR-27:** The owner shall be able to soft-remove an active Attachment after confirmation and entry of a valid reason.
- **FR-28:** A removed Attachment shall retain metadata but shall not be previewable, downloadable, or removable again.
- **FR-29:** Direct operations on another Requester's Attachment shall return no metadata, bytes, or mutation.

Presentation and Feedback

- **FR-30:** All screens shall follow `ui-spec.md`, be keyboard-operable, and work at desktop, tablet, and mobile widths without page-level horizontal scrolling.
- **FR-31:** All asynchronous actions shall have visible loading/busy feedback and prevent repeated activation.
- **FR-32:** User-facing errors shall be safe and actionable; internal exception details shall remain server-side.

5. Business Rules

Identity and Ownership

- **BR-01:** The official Ticket Number is generated by the backend and must be unique.
- **BR-02:** A new Ticket begins with Current Status 'NEW'.
- **BR-03:** Development Requester Selection is a Lab 2 testing mechanism and is not authentication.
- **BR-04:** Only active Requesters may be selected or create new Tickets.
- **BR-05:** The frontend sends the temporary selected identity through `X-Requester-Id`; the backend validates that it identifies an active Requester.
- **BR-06:** `requesterId` is not an editable Ticket field. The backend derives ownership from the validated Requester context.
- **BR-07:** A Ticket or Attachment may be accessed only when owned by the selected Requester.

- **BR-08:** Missing and non-owned Ticket/Attachment resources use the same safe `404` response to avoid exposing another Requester's resource existence.
- **BR-09:** Changing Requester clears owner-specific cached results and confirms before discarding a dirty Create Ticket form.
- **BR-10:** A saved selection that no longer identifies an active Requester is cleared and the selector is shown.

Ticket Values and Validation

- **BR-11:** Ticket Number format is `TKT-YYYY-NNNNNN`; `YYYY` is the server UTC creation year and `NNNNNN` is a zero-padded PostgreSQL sequence value. Sequence gaps are permitted.
- **BR-12:** Ticket Date is the backend UTC `createdAt`; the UI may format it in the user's local timezone.
- **BR-13:** Category and Related System must exist and be active at submission time.
- **BR-14:** Requested Priority is one of `LOW`, `MEDIUM`, `HIGH`, or `URGENT`; the UI defaults to `MEDIUM`.
- **BR-15:** Summary is required, trimmed, and 5-120 characters after trimming.
- **BR-16:** Description is required, trimmed, and 10-5,000 characters after trimming.
- **BR-17:** Whitespace-only values are invalid. Internal whitespace and line breaks are preserved.
- **BR-18:** Backend response/database values are authoritative for Ticket Number, Ticket Date, Requester, status, and saved fields.
- **BR-19:** Each create attempt contains a UUID `clientRequestId`; `(requesterId, clientRequestId)` is unique.
- **BR-20:** A replay with the same Requester, request ID, and normalized payload returns the existing Ticket without duplication. Reusing the ID for different content returns `409`.

Search, Filter, Sort, and Pagination

- **BR-21:** Search is case-insensitive and matches Ticket Number or Summary after trimming.
- **BR-22:** Search and active filters are combined using logical AND.
- **BR-23:** Supported filters are Category ID, Related System ID, Requested Priority, and Current Status.
- **BR-24:** Supported sort fields are `createdAt`, `updatedAt`, and `ticketNumber`; directions are `asc` and `desc`.
- **BR-25:** Default order is `updatedAt desc`, then `id desc`; other sorts also use a stable ID tie-breaker.
- **BR-26:** Pagination is one-based. Default page size is 10; allowed sizes are 10, 20, and 50.
- **BR-27:** Invalid query values return `400`. A valid page beyond the final page returns an empty array with accurate metadata.
- **BR-28:** Empty state means the Requester owns no Tickets; no-results means current search/filters match none.

Attachment Rules

- **BR-29:** Allowed types are JPG/JPEG, PNG, WEBP, and PDF.
- **BR-30:** Maximum size is 5 MiB (5,242,880 bytes) per file.
- **BR-31:** A Ticket has at most five active Attachments; removed records do not count toward the limit.
- **BR-32:** Acceptance requires an allowed extension, MIME type, and matching file signature.
- **BR-33:** Original filenames are sanitized for display: path separators, control characters, and reserved filename characters are replaced with `\_`, and the result is truncated to 255 Unicode code points while preserving its extension. Storage names are generated UUIDs and never use the client path.
- **BR-34:** Files are stored outside the public web root; download always passes through ownership and active-state checks.
- **BR-35:** Upload accepts one file per request. Any file written before a metadata failure is removed as compensation.
- **BR-36:** Removal is soft removal using `removedAt`, `removedReason`, and `removedByRequesterId`.
- **BR-37:** Removal reason is required, trimmed, and 5-250 characters.
- **BR-38:** Removed metadata retains filename, type, size, upload time, removal time, and reason, but has no preview/download action.
- **BR-39:** Removed or non-owned downloads return the safe `404` response.
- **BR-40:** If Ticket creation succeeds but an Attachment upload fails, the Ticket remains saved and the UI offers a retry path.

Error and State Rules

- **BR-41:** Validation errors use the shared safe API error envelope; stack traces, SQL details, filesystem paths, and secrets are never returned.
- **BR-42:** Failed Ticket creation preserves editable form values; a successful response clears those values only after confirmation is shown.
- **BR-43:** Busy controls prevent repeated UI activation but do not replace backend idempotency and validation.
- **BR-44:** Reference data that becomes inactive after page load is rejected by the backend and reloaded by the UI.
- **BR-45:** Lab 3 may replace the temporary Requester header with authenticated identity while preserving `Ticket.requesterId` ownership.

6. UI Specification Summary

- Use the Zen Green tokens and reusable shell defined in `ui-spec.md`.
- Clearly label the Requester selector as testing-only and not login.
- Distinguish editable, read-only, invalid, disabled, focused, and busy controls.
- Use a desktop Ticket table and equivalent mobile cards.
- Keep Ticket Detail read-only and separate Ticket information from Attachment actions.
- Use text-labelled Requested Priority and Current Status badges.
- Do not implement IT Priority controls; only reserve reusable future styling if needed.
- Never rely on color alone for meaning.

7. Data Changes

Models

Model	Required fields and constraints
---	---
`RequesterUser`	`id` integer PK; `name` max 120; unique normalized `email` max 254; `isActive` default true; `createdAt`; `updatedAt`
`Category`	Existing `id`, unique `name`, `createdAt`; add `isActive` default true and `updatedAt`
`RelatedSystem`	`id` integer PK; unique `name`; `isActive` default true; `createdAt`; `updatedAt`
`Ticket`	`id`; unique non-null `ticketNumber`; owner/category/system FKs; `summary`; `description`; priority/status enums; `clientRequestId`; `requestPayloadHash`; timestamps; unique `(requesterId, clientRequestId)`
`Attachment`	`id`; `ticketId` FK; `originalName`; unique `storageKey`; `mimeType`; `sizeBytes`; `uploadedAt`; nullable removal fields and remover FK

Enums

- `RequestedPriority`: `LOW`, `MEDIUM`, `HIGH`, `URGENT`.
- `TicketStatus`: `NEW` in Lab 2, designed for explicit future migrations.

Relationships

- One `RequesterUser` owns many Tickets; each Ticket has exactly one Requester.
- One Category and one Related System may relate to many Tickets.
- One Ticket has many Attachments.
- One Requester may be recorded as remover of many Attachments.

Indexes and Constraints

- Unique: Requester email, Category name, Related System name, Ticket Number, Attachment storage key, and `(requesterId, clientRequestId)`.
- Foreign-key indexes for all relationships.
- Default-list index `(requesterId, updatedAt desc, id desc)`.
- Owner-prefixed filter indexes for Category, Related System, Requested Priority, and Current Status.
- Removal fields must be all null or all populated through service validation and a database check constraint where appropriate.

Migration and Seed

- Preserve all Lab 1 Category IDs and data; new Category columns receive safe defaults.
- Add a PostgreSQL sequence for Ticket Numbers.
- Seed with `upsert` so repeat execution creates no duplicates.
- Include Categories: Account and Access, Hardware, Software, Network.
- Include at least seven Related Systems: Email, Campus Wi-Fi, VPN, LEB2 App, Grade Submission App, Printer, Corporate Laptop.
- Include at least four active and one inactive Development Requester.
- Never commit `.env`, credentials, or uploaded files.

Justified Database Decision

Ticket ownership is stored as a direct foreign key to `RequesterUser`, not inferred from browser state. This supports indexed owner-scoped queries and enforceable backend checks in Lab 2. Lab 3 can later associate an authenticated account with the same Requester record without rewriting existing Ticket ownership.

8. API Contract

The normative contract is `api-spec.md`. It defines active reference-data retrieval, idempotent Ticket creation, owner-scoped list/detail retrieval, Attachment lifecycle, request/response shapes, validation, pagination, status codes, safe errors, and ownership behavior.

9. Acceptance Criteria

- ****AC-01:**** Given no valid selected Requester, when a requester-specific route opens, then the selector is shown and protected content is not rendered.
- ****AC-02:**** Given active and inactive seeded Requesters, when the selector loads, then only active Requesters are offered.
- ****AC-03:**** Given the Requester API is loading, empty, or failed, when the selector renders, then the matching accessible state appears.
- ****AC-04:**** Given Requester A is selected, when Continue is activated, then the shell displays A and requester-specific requests use A's context.
- ****AC-05:**** Given A is current, when the user changes to B, then A-owned client data is cleared and B-owned data loads.
- ****AC-06:**** Given Create Ticket opens, when reference data loads, then active Categories/Systems come from the API and Requester is read-only from context.
- ****AC-07:**** Given invalid or boundary Ticket input, when submission is attempted, then field errors appear and no invalid Ticket is saved.
- ****AC-08:**** Given valid Ticket data, when submitted, then exactly one Ticket is saved for the selected Requester with backend Ticket Number, date, and `NEW` status.
- ****AC-09:**** Given a successful request is retried with the same request ID/payload, when received, then no duplicate is created and the original Ticket is returned.
- ****AC-10:**** Given submission is in progress, when repeated activation is attempted, then the UI remains busy/disabled and backend idempotency prevents duplication.
- ****AC-11:**** Given a create API failure, when handled, then a safe error appears and editable values remain available.
- ****AC-12:**** Given a Ticket is created and an Attachment upload fails, when results appear, then the Ticket remains saved and failed files can be retried.
- ****AC-13:**** Given Requester A owns Tickets, when My Tickets loads as A, then only A's Tickets are returned.
- ****AC-14:**** Given matching/nonmatching Tickets, when search is used, then Ticket Number/Summary matching is case-insensitive and no-results appears when appropriate.
- ****AC-15:**** Given valid filter combinations, when applied, then search and all filters use AND semantics.
- ****AC-16:**** Given a supported sort, when applied, then results use that order with deterministic tie-breaking.
- ****AC-17:**** Given multiple pages, when pagination/page size changes, then the correct owner-scoped slice and accurate metadata are returned.
- ****AC-18:**** Given invalid query parameters, when requested, then the API returns safe `400` and the UI offers recovery.
- ****AC-19:**** Given an owner with no Tickets versus filters with no matches, when rendered, then distinct empty and no-results states appear.
- ****AC-20:**** Given A owns a Ticket, when A opens it, then approved Ticket fields and Attachment metadata render read-only.
- ****AC-21:**** Given B directly requests A's Ticket, when ownership is checked, then safe `404` is returned with no A data.
- ****AC-22:**** Given an allowed file at or below 5 MiB and fewer than five active files, when the owner uploads it, then active metadata and storage are created.
- ****AC-23:**** Given invalid type/signature, oversize, or a sixth active file, when uploaded, then it is rejected without active metadata.
- ****AC-24:**** Given an active owned Attachment, when downloaded, then correct bytes, media type, and safe filename are returned.
- ****AC-25:**** Given an active owned Attachment and valid reason, when removal is confirmed, then removal metadata is stored and active count decreases.
- ****AC-26:**** Given a removed Attachment, when viewed, then retained metadata/reason appear but download/preview/remove actions do not.
- ****AC-27:**** Given a removed or non-owned Attachment, when operated on directly, then safe rejection returns no bytes or mutation.
- ****AC-28:**** Given keyboard-only use, when navigating each screen, then controls, focus, labels, errors, and dialogs are understandable without color alone.
- ****AC-29:**** Given desktop, tablet, and mobile viewports, when required screens/states are inspected, then no clipping, overlap, hidden action, or page-level horizontal scroll exists.
- ****AC-30:**** Given final `main`, when documented build/test commands run, then every required non-skipped test passes and every AC maps to evidence.

10. Definition of Done

Product Completion

- [x] All approved FRs/BRs are implemented without excluded Lab 3 or IT Staff features.
- [x] Every AC maps to one or more passing tests.

- [x] Required unit, API, UI, style, responsive, visual, and E2E evidence passes on final `main`.
- [x] No required test is skipped, disabled, commented out, flaky, or unrelated.
- [x] Prisma schema, migrations, constraints, indexes, and idempotent seed match the contract.
- [x] API behavior matches `api-spec.md` and UI matches `ui-spec.md`.
- [x] Success, invalid, boundary, loading, empty, no-results, ownership, storage, and failure cases are demonstrated.
- [x] README setup, migration, seed, run, test, and storage instructions are current.
- [x] No secrets, `.env`, generated build output, or uploaded files are committed.

Course Delivery

- [x] GitHub Issues cover specifications, data, API, UI, tests, E2E, visual review, documents, and release.
- [x] `lab2-staging` was created from completed Lab 1 `main`.
- [x] Every Issue uses a feature branch and peer-reviewed PR into `lab2-staging`.
- [x] Real reviewer comments, responses, corrections, and approvals are recorded.
- [x] Release PR `lab2-staging -> main` passes integration tests and review.
- [x] `ai-use.md` contains 6-10 actual prompts and the student's own reflection.
- [x] Required screenshots are readable under `artifacts/lab-02/screenshots/`.
- [x] One concise PDF uses `Answer Part 1` through `Answer Part 9` in exact order with working links.

11. Assumptions and Decisions Requiring Human Approval

****Human review record (2026-08-24):**** The student reviewed and approved the Engineering Contract as written, including the Functional Requirements, Business Rules, Acceptance Criteria, Test Plan, UI Specification, API contract, and decisions D-01 through D-13.

ID	Proposed decision	Reason	Human decision
---	---	---	---
D-01	Store selected Requester ID in `sessionStorage`	Simulates one tab/session without pretending to be authentication.	Approved
D-02	Send temporary context through `X-Requester-Id`; never accept editable body ownership	Consistent owner context and no body/header mismatch.	Approved
D-03	Return safe `404` for missing and non-owned Ticket/Attachment	Avoids resource enumeration.	Approved
D-04	Use `TKT-YYYY-NNNNNN` from PostgreSQL sequence	Backend-generated, concurrent, unique, readable, testable.	Approved
D-05	Summary 5-120; Description 10-5,000; removal reason 5-250	Bounded payloads with usable content.	Approved
D-06	Priorities `LOW`, `MEDIUM`, `HIGH`, `URGENT`; UI default `MEDIUM`	Clear Requester choice and future-compatible enum.	Approved
D-07	UUID request ID plus unique database constraint/replay	Covers double-clicks, retries, and concurrent duplicates.	Approved
D-08	Search Ticket Number and Summary only	Meets discoverability without broad Description search.	Approved
D-09	One-based pagination; 10 default; 10/20/50 allowed	Bounded, simple list contract.	Approved
D-10	Upload one Attachment per API request after Ticket creation	Clear per-file outcome and retryable partial failure.	Approved
D-11	Validate extension, MIME, and signature; exact 5 MiB boundary	Prevents simple renamed-file bypasses.	Approved
D-12	Retain removed binary outside public access but block application download/preview	Preserves soft-removal metadata and required unavailable state.	Approved
D-13	No Lab 2 IT Priority workflow; reserve future badge styling only	Avoids excluded IT Staff scope.	Approved

Answer Part 3: Test-Driven Development and Traceability

The test plan maps every Acceptance Criterion to unit, API/integration, UI, responsive, visual, and E2E evidence. The final rendered test plan includes actual test-file paths, final status, commands, and the completed visual checklist.

Rendered test plan: <docs/lab-02/tests.md>

Final command output from main:

```
Server: 11 test files, 62 tests passed (RUN_DB_INTEGRATION=1)
Client: 9 test files, 54 tests passed
Playwright: 9 tests passed
Prisma migrate deploy: no pending migrations
Seed: two consecutive successful runs with idempotent data
```

Traceability: [Acceptance Criteria and test matrix](#)

Rendered copy: docs/lab-02/tests.md

[docs/lab-02/tests.md](#)

Lab 2 Test Plan and Results

1. Test Strategy

- **Unit:** formatting, normalization, query parsing, file detection, and business-rule helpers.
- **API/integration:** Express + Prisma/PostgreSQL contracts, database constraints, ownership, validation, failure compensation, and safe errors.
- **UI component:** React behavior using mocked API boundaries, including validation and asynchronous states.
- **UI style/accessibility:** tokens/classes, labels, field states, focus, keyboard behavior, and automated checks where practical.
- **Responsive/visual:** Playwright viewports and Human inspection against `ui-spec.md`.
- **E2E:** real client, server, PostgreSQL, migrations, seed, isolated storage, and complete Requester workflows.

Database tests use a dedicated test database/schema. Attachment tests use a task-specific temporary storage directory and known fixtures; production/user uploads are never used.

TDD Cycle for Every Implementation Issue

1. Select mapped FR/BR/AC and planned tests.
2. Add the smallest failing tests.
3. Run them and confirm they fail for the expected missing behavior.
4. Implement only the current Issue's scope.
5. Run focused tests until green.
6. Refactor while tests remain green.
7. Run relevant regression suites.
8. Record final results from actual commands only.

2. Planned Unit Tests

ID	Requirement / AC	What it tests	Expected result	Test file	Final
---	---	---	---	---	---
UNIT-01	BR-11, AC-08	Ticket Number formatting	Exact `TKT-YYYY-NNNNNN` with padding	`server/tests/lab-02/create-ticket.api.test.ts`, `create-ticket.postgres.integration.test.ts`	PASS
UNIT-02	BR-15-BR-17, AC-07	Summary/Description trimming and boundaries	Exact limits pass; outside/blank fail	`server/tests/lab-02/create-ticket.api.test.ts`	PASS
UNIT-03	BR-19-BR-20, AC-09	Normalized payload hash/idempotency	Equivalent content hashes equally; changes differ	`server/tests/lab-02/create-ticket.api.test.ts`, `create-ticket.postgres.integration.test.ts`	PASS
UNIT-04	BR-21-BR-27, AC-14-AC-18	Ticket-list query parsing/defaults	Valid normalized; invalid/unknown rejected	`server/tests/lab-02/my-tickets.api.test.ts`	PASS
UNIT-05	BR-29-BR-32, AC-22-AC-23	Extension/MIME/signature and 5 MiB boundary	Valid accepted; mismatch/oversize rejected	`server/tests/lab-02/attachments.api.test.ts`, `attachments.postgres.integration.test.ts`	PASS
UNIT-06	BR-33-BR-34, AC-22	Filename sanitization/storage key	No traversal/control chars; generated UUID key	`server/tests/lab-02/attachments.api.test.ts`	PASS

3. Planned API and Integration Tests

ID	Requirement / AC	What it tests	Expected result	Test file	Final
---	---	---	---	---	---
API-01	FR-02, BR-04, AC-02	Active Requester list	`200`, active only, deterministic order	`server/tests/lab-02/reference-data.api.test.ts`	PASS
API-02	FR-02, AC-03	Empty/failing Requester API	Empty `200 []`; safe `500`	`server/tests/lab-02/reference-data.api.test.ts`	PASS
API-03	BR-05, AC-01, AC-04	Missing/malformed/inactive/valid context	Invalid `400`; valid accepted	`server/tests/lab-02/reference-data.api.test.ts`, `create-ticket.api.test.ts`	PASS
API-04	FR-08, BR-13, AC-06	Active Categories/Related Systems	Correct shape, active only, deterministic	`server/tests/lab-02/reference-data.api.test.ts`	PASS
API-05	FR-11, BR-01-BR-02, AC-08	Valid Ticket creation	`201`; one row; owner/number/`NEW` correct	`server/tests/lab-02/create-ticket.api.test.ts`, `create-ticket.postgres.integration.test.ts`	PASS
API-06	FR-10, BR-14-BR-17, AC-07	Required fields, enum, trim, boundaries	`400` field errors; no row	`server/tests/lab-02/create-ticket.api.test.ts`	PASS

API-07 BR-06, AC-08 Body attempts owner/system override `400`; no generated/owner override `server/tests/lab-02/create-ticket.api.test.ts` PASS
API-08 BR-13, BR-44, AC-07 Missing/inactive Category/System `400`; no Ticket `server/tests/lab-02/create-ticket.api.test.ts` PASS
API-09 FR-12, BR-19-BR-20, AC-09-AC-10 Same request ID/payload replay First `201`, replay `200`, one Ticket `server/tests/lab-02/create-ticket.api.test.ts`, `create-ticket.postgres.integration.test.ts` PASS
API-10 BR-20, AC-09 Same request ID/different payload `409`; original unchanged `server/tests/lab-02/create-ticket.api.test.ts` PASS
API-11 BR-01, AC-08 Concurrent Ticket creation All committed Ticket Numbers unique `server/tests/lab-02/create-ticket.postgres.integration.test.ts` PASS
API-12 FR-13, BR-41, AC-11 Simulated database failure Safe `500`; no partial Ticket `server/tests/lab-02/create-ticket.api.test.ts` PASS
API-13 FR-15, BR-07, AC-13 A/B list isolation Each response contains its owner only `server/tests/lab-02/my-tickets.api.test.ts` PASS
API-14 FR-16, BR-21, AC-14 Case-insensitive Ticket Number/Summary search Correct owner-scoped matches `server/tests/lab-02/my-tickets.api.test.ts` PASS
API-15 FR-17, BR-22-BR-23, AC-15 Combined search/filters AND semantics `server/tests/lab-02/my-tickets.api.test.ts` PASS
API-16 FR-18, BR-24-BR-25, AC-16 Supported sorts/tie ordering Deterministic exact order `server/tests/lab-02/my-tickets.api.test.ts` PASS
API-17 FR-19, BR-26, AC-17 Page boundaries and sizes 10/20/50 Correct slice/metadata `server/tests/lab-02/my-tickets.api.test.ts` PASS
API-18 BR-27-BR-28, AC-17, AC-19 Page beyond final/owner empty `200` empty with accurate totals `server/tests/lab-02/my-tickets.api.test.ts` PASS
API-19 FR-20, BR-27, AC-18 Invalid/unknown query Safe `400 INVALID_QUERY` `server/tests/lab-02/my-tickets.api.test.ts` PASS
API-20 FR-20, BR-41, AC-18 Ticket-list database failure Safe `500` with no internals `server/tests/lab-02/my-tickets.api.test.ts` PASS
API-21 FR-22, AC-20 Owned Ticket Detail `200`; approved detail/metadata shape `server/tests/lab-02/ticket-detail.api.test.ts` PASS
API-22 FR-23, BR-07-BR-08, AC-21 Missing/cross-owner Ticket Same safe `404`; no owner data `server/tests/lab-02/ticket-detail.api.test.ts` PASS
API-23 FR-25, BR-29-BR-35, AC-22 Valid types and exact 5 MiB upload `201`; one active row/file `server/tests/lab-02/attachments.api.test.ts`, `attachments.postgres.integration.test.ts` PASS
API-24 BR-29-BR-32, AC-23 Type/signature mismatch and >5 MiB `415`/`413`; no row/file `server/tests/lab-02/attachments.api.test.ts` PASS
API-25 BR-31, AC-23 Fifth/sixth active Attachment Fifth accepted; sixth `409`; removed excluded `server/tests/lab-02/attachments.api.test.ts`, `attachments.postgres.integration.test.ts` PASS
API-26 BR-35, AC-22 Storage/metadata compensation Safe `500`; no orphan/active row `server/tests/lab-02/attachments.api.test.ts`, `attachments.postgres.integration.test.ts` PASS
API-27 FR-26, BR-34, AC-24 Active owned download Correct bytes/MIME/name/`nosniff` `server/tests/lab-02/attachments.api.test.ts`, `attachments.postgres.integration.test.ts` PASS
API-28 FR-27, BR-36-BR-38, AC-25-AC-26 Valid soft removal `200`; metadata retained; count decreases `server/tests/lab-02/attachments.api.test.ts`, `attachments.postgres.integration.test.ts` PASS
API-29 BR-37, AC-25 Removal-reason boundaries Invalid `400`; 5/250 accepted; >250 rejected `server/tests/lab-02/attachments.api.test.ts` PASS
API-30 FR-28-FR-29, BR-39, AC-26-AC-27 Removed/non-owned/wrong-Ticket access Safe `404`; no bytes/mutation `server/tests/lab-02/attachments.api.test.ts`, `attachments.postgres.integration.test.ts` PASS
API-31 FR-24, BR-38, AC-26 Metadata after removal Active/removed states; removed URL null `server/tests/lab-02/attachments.api.test.ts`, `attachments.postgres.integration.test.ts` PASS

4. Planned UI, Style, and Accessibility Tests

ID	Requirement / AC	What it tests	Expected result	Test file	Final
---	---	---	---	---	---
UI-01	FR-01, AC-01	Route guard without Requester	Selector shown; protected screen absent	`client/tests/lab-02/DevelopmentRequesterSelect.test.tsx`	PASS
UI-02	FR-02-FR-03, AC-02-AC-03	Selector ready/loading/empty/failure	Correct accessible states/Retry	`client/tests/lab-02/DevelopmentRequesterSelect.test.tsx`	PASS
UI-03	FR-04-FR-06, AC-04-AC-05	Restore/current identity/Change Requester	Header updates; A state cleared; B is revalidated and rendered	`client/tests/lab-02/DevelopmentRequesterSelect.test.tsx`	PASS
UI-04	FR-07-FR-09, AC-06	Create initial/reference/read-only fields	API data, current Requester, pending values	`client/tests/lab-02/CreateTicket.test.tsx`	PASS
UI-05	FR-10, BR-14-BR-17, AC-07	Field validation/focus	Near-field errors; no API call; focus first invalid	`client/tests/lab-02/CreateTicket.test.tsx`	PASS
UI-06	FR-11, AC-08	Successful creation	Backend Ticket Number/saved values/next actions	`client/tests/lab-02/CreateTicket.test.tsx`	PASS
UI-07	FR-12, AC-10	Busy/disabled Submit	One request; visible busy text	`client/tests/lab-02/CreateTicket.test.tsx`	PASS
UI-08	FR-13, AC-11	Create API failure	Safe alert; values/files retained; Retry	`client/tests/lab-02/CreateTicket.test.tsx`	PASS
UI-09	FR-14, AC-12	Partial Attachment failure	Ticket success retained; failed file/retry shown	`client/tests/lab-02/CreateTicket.test.tsx`	PASS

UI-10 BR-29-BR-32, AC-23 Invalid/sixth selected file Per-file reason; invalid not submitted `client/tests/lab-02/CreateTicket.test.tsx` PASS
UI-11 FR-15, AC-13 My Tickets owner switch A disappears; loading then B appears `client/tests/lab-02/MyTickets.test.tsx` PASS
UI-12 FR-16-FR-17, AC-14-AC-15 Search/filters/Clear Correct query/page reset/preserved controls `client/tests/lab-02/MyTickets.test.tsx` PASS
UI-13 FR-18-FR-19, AC-16-AC-17 Sort/page/page size Correct query and control states `client/tests/lab-02/MyTickets.test.tsx` PASS
UI-14 FR-20, AC-18-AC-19 Loading/empty/no-results/query/failure Distinct messages/actions `client/tests/lab-02/MyTickets.test.tsx` PASS
UI-15 FR-21, AC-20 Open Ticket Detail View Ticket opens the owned Ticket Detail screen `client/tests/lab-02/MyTickets.test.tsx`, `client/tests/lab-02/RequesterTicketDetail.test.tsx` PASS
UI-16 FR-22-FR-24, AC-20-AC-21 Read-only detail/denied state Approved read-only fields and active/removed Attachment metadata, or safe failure `client/tests/lab-02/RequesterTicketDetail.test.tsx` PASS
UI-17 FR-25-FR-26, AC-22-AC-24 Upload/download states Busy/success/error/active controls `client/tests/lab-02/AttachmentSection.test.tsx` PASS
UI-18 FR-27-FR-28, AC-25-AC-26 Removal dialog/reason/removed state Accessible confirm; metadata; actions absent `client/tests/lab-02/AttachmentSection.test.tsx` PASS
UI-19 FR-29, AC-27 Removed/unauthorized/unavailable errors Safe state; no file/action exposure `client/tests/lab-02/AttachmentSection.test.tsx` PASS
UI-20 FR-30-FR-32, AC-28 Labels/markers/field/button states Required semantics/classes/text indicators `client/tests/lab-02/ZenGreenStyle.test.tsx` PASS
UI-21 AC-28 Keyboard and semantic accessibility checks Logical focus, labels, announcements, and text states `client/tests/lab-02/accessibility.test.tsx` PASS

5. Planned E2E, Responsive, and Visual Tests

ID	Requirement / AC	What it tests	Expected result	Test file	Final
---	---	---	---	---	---
E2E-01	AC-04, AC-06-AC-12	Select Requester, create Ticket, mixed files, simulated failure	Backend values, one Ticket, documented recovery	`client/e2e/lab-02/requester-ticket-flow.spec.ts`	PASS
E2E-02	AC-13-AC-21	Create as A, list controls, switch to B, direct A detail A/B isolation and safe rejection		`client/e2e/lab-02/requester-ticket-flow.spec.ts`	PASS
E2E-03	AC-22-AC-27	Add, download, remove, retained metadata, blocked retry	Complete Attachment lifecycle	`client/e2e/lab-02/requester-ticket-flow.spec.ts`	PASS
E2E-04	AC-29	Desktop 1440x900, tablet 820x1180, mobile 390x844	No clipping/overlap/page scroll; usable controls	`client/e2e/lab-02/responsive-visual.spec.ts`	PASS
E2E-05	AC-29	Required screenshots and Human checklist	Artifacts stored and Human-approved	`client/e2e/lab-02/visual-evidence.spec.ts`	PASS
E2E-06	AC-30	Full builds/tests/seed/workflow on final main	All required commands pass, no skips	`client/e2e/lab-02/requester-ticket-flow.spec.ts`, `client/e2e/lab-02/visual-evidence.spec.ts`	PASS

6. Acceptance-Criterion Traceability

AC	Planned evidence
---	---
AC-01	API-03, UI-01
AC-02	API-01, UI-02
AC-03	API-02, UI-02
AC-04	API-03, UI-03, E2E-01
AC-05	UI-03, UI-11, E2E-02
AC-06	API-04, UI-04, E2E-01
AC-07	UNIT-02, API-06-API-08, UI-05
AC-08	UNIT-01, API-05, API-07, API-11, UI-06, E2E-01
AC-09	UNIT-03, API-09-API-10
AC-10	API-09, UI-07, E2E-01
AC-11	API-12, UI-08, E2E-01
AC-12	UI-09, E2E-01
AC-13	API-13, UI-11, E2E-02
AC-14	UNIT-04, API-14, UI-12, E2E-02
AC-15	API-15, UI-12, E2E-02
AC-16	API-16, UI-13, E2E-02
AC-17	API-17-API-18, UI-13, E2E-02
AC-18	UNIT-04, API-19-API-20, UI-14
AC-19	API-18, UI-14, E2E-02

AC-20 API-21, UI-15-UI-16, E2E-02
AC-21 API-22, UI-16, E2E-02
AC-22 UNIT-05-UNIT-06, API-23, API-26, UI-17, E2E-03
AC-23 UNIT-05, API-24-API-25, UI-10, E2E-01
AC-24 API-27, UI-17, E2E-03
AC-25 API-28-API-29, UI-18, E2E-03
AC-26 API-28, API-31, UI-18, E2E-03
AC-27 API-30, UI-19, E2E-03
AC-28 UI-20-UI-21
AC-29 E2E-04-E2E-05 and Human visual checklist
AC-30 E2E-06 and final command/traceability audit

7. Responsive and Visual Checklist

- [x] Desktop `1440x900`: centered max-width, multi-column form, full Ticket table.
- [x] Tablet `820x1180`: two columns where practical; long fields/Attachments have sufficient width.
- [x] Mobile `390x844`: stacked form, Ticket cards, touch-friendly actions, no page-level horizontal scroll.
- [x] No clipped labels, messages, badges, filenames, pagination, or navigation.
- [x] No overlapping dialogs, buttons, cards, or error messages.
- [x] Editable/read-only/invalid/disabled/focused controls match `ui-spec.md`.
- [x] Button hierarchy and busy/destructive states are consistent.
- [x] Empty and no-results states are distinct and actionable.
- [x] Desktop table/mobile cards contain equivalent information.
- [x] Attachment states are clear.
- [x] Keyboard focus is visible and logical.
- [x] No meaning relies only on color.

8. Planned Test Commands

Commands must be updated only if the implemented tooling requires a documented change.

```
# From repository root
npm --prefix server run build
npm --prefix server test
npm --prefix client run build
npm --prefix client test

# Database, from server/
Set-Location server
npx prisma migrate deploy
npm run prisma:seed
npm run prisma:seed
Set-Location ..

# Playwright configured through client/
Set-Location client
npm run e2e
Set-Location ..
```

Focused commands belong in Issue/PR evidence; final evidence must come from complete commands on final `main`.

9. Current and Final Results

Lab 1 Baseline

- Lab 1 documentation reports Server: 2 files/2 tests passed.
- Lab 1 documentation reports Client: 1 file/3 tests passed.
- These results must be rerun as regression evidence when Lab 2 implementation begins.

Lab 2

****Status:**** Feature 17 Zen Green style and accessibility evidence is implemented on branch `feature/17-lab2-style-accessibility`; UI-20/UI-21 coverage, keyboard semantics, responsive card/table behavior, and reduced-motion behavior are included.

Feature 10 Evidence

- Implementation and integration-test commit: `3e8e6c2`.
- `npm test` with `RUN\_DB\_INTEGRATION=1`: **42 tests passed**.
- `npm run build`: **passed**.
- PostgreSQL integration covers owned detail, A/B owner isolation, active/removed Attachment ordering, and safe 404 behavior.
- Hosted evidence is provided by GitHub Actions workflow `Server CI` in `.github/workflows/server-ci.yml`; the post-fix run will execute on PR #23 after this branch is pushed.

Feature 12 Evidence

- Client suite: **11 tests passed total** = 6 Feature 12 requester-selection tests + 2 Requester API shape tests + 3 Lab 1 regression tests updated for the Lab 2 route guard.
- `npm run build` from `client/`: **passed**.
- Coverage includes route guard, active Requester loading/empty/failure states, disabled Continue, response-shape validation, sessionStorage restore, current identity, Change Requester ADB revalidation, and unavailable navigation placeholders.

Feature 13 Evidence

- Client suite: **24 tests passed total** = 13 Feature 13 Create Ticket tests + 6 Feature 12 requester-selection tests + 2 Requester API shape tests + 3 Lab 1 regression tests.
- `npm test` from `client/`: **passed** (4 test files, 24 tests).
- `npm run build` from `client/`: **passed**.
- Coverage includes active Category/Related System loading, read-only Ticket fields, default Medium priority, field-level validation and first-invalid focus, authoritative Ticket Number, busy/duplicate-submit protection, stable clientRequestId across create retry, fresh clientRequestId for a new Ticket, safe create failure with retained files, cumulative picker/quota behavior, extension/MIME validation, per-file Attachment validation/removal, invalid-file exclusion, and individual retry for partial Attachment failure.

Feature 14 Evidence

- Client suite: **31 tests passed total** = 7 Feature 14 My Tickets tests + 13 Feature 13 Create Ticket tests + 6 Feature 12 requester-selection tests + 2 Requester API shape tests + 3 Lab 1 regression tests.
- `npm test` from `client/`: **passed** (5 test files, 31 tests).
- `npm run build` from `client/`: **passed**.
- Coverage includes owner-scoped Ticket loading, ADB requester switching without stale data, semantic desktop table and mobile card rendering, `aria-sort` state, labelled loading/no-stale-data state, search/filter AND query reset, clear filters, owner-empty and filtered no-results states, safe retry, page size, and pagination controls.

Feature 15 Evidence

- Client suite: **36 tests passed total** = 5 Feature 15 Ticket Detail tests + 7 Feature 14 My Tickets tests + 13 Feature 13 Create Ticket tests + 6 Feature 12 requester-selection tests + 2 Requester API shape tests + 3 Lab 1 regression tests.
- `npm test` from `client/`: **passed** (6 test files, 36 tests).
- `npm run build` from `client/`: **passed**.
- Coverage includes View Ticket navigation, owned read-only Ticket fields, Requester context, loading/no-stale-data, safe failure/retry, safe 404 not-found handling without Ticket-data leakage, active/removed Attachment metadata with removal timestamp, no download action before Attachment UI, and back-navigation preserving non-default My Tickets query values.

Feature 16 Evidence

- Client suite: **45 tests passed total** = 9 Feature 16 Attachment UI tests + 5 Feature 15 Ticket Detail tests + 7 Feature 14 My Tickets tests + 13 Feature 13 Create Ticket tests + 6 Feature 12 requester-selection tests + 2 Requester API shape tests + 3 Lab 1 regression tests.
- `npm test` from `client/`: **passed** (7 test files, 45 tests).
- `npm run build` from `client/`: **passed**.
- Coverage includes active Attachment download with requester context, per-file upload success/failure/retry, cumulative client validation for MIME/type/duplicate files, same-name different-content acceptance, real sixth-file quota visibility, focus trap/Escape/trigger-focus restoration, Removing busy state and duplicate activation prevention, accessible removal dialog with 5-250 character reason validation, removed Attachment action suppression, and safe storage-unavailable messaging.

Feature 17 Evidence

- Client suite: **54 tests passed total** = 9 Feature 17 Zen Green style/accessibility tests + 9 Feature 16 Attachment UI tests + 5 Feature 15 Ticket Detail tests + 7 Feature 14 My Tickets tests + 13 Feature 13 Create Ticket tests + 6 Feature 12 requester-selection tests + 2 Requester API shape tests + 3 Lab 1 regression tests.

- `npm test` from `client/`: **passed** (9 test files, 54 tests).
- `npm run build` from `client/`: **passed**.
- Coverage includes required markers and programmatic labels, read-only/editable distinction, primary/secondary/busy button states, equivalent desktop table/mobile card content, semantic sort state, active navigation `aria-current`, first-invalid focus, loading `aria-busy`/status announcements, text-labelled priority/status/actions, focus-visible styles, and reduced-motion support.

Feature 18 Evidence

- Scope is Issue #38 / E2E-01 only; E2E-02 through E2E-06 remain planned for the subsequent ownership, Attachment, responsive, visual-evidence, and final-release Issues.
- Playwright E2E suite: **2 tests passed** for E2E-01.
- `npm run e2e` from `client/`: **passed** (2 tests, Chromium; real Vite client, Express API, PostgreSQL, migrations, seed, and Attachment storage).
- Hosted E2E CI runs the Prisma seed twice before Playwright to verify repeatable reference-data setup.
- Coverage includes active Requester selection, real reference-data loading, valid Ticket creation with mixed PDF/PNG Attachments, backend-generated Ticket Number and `NEW` status, simulated ambiguous create response with form retention, idempotent retry without duplicate Ticket, partial Attachment failure, and individual Attachment retry.

Feature 19 Evidence

- Scope is Issue #39 / E2E-02 only; E2E-03 through E2E-06 remain planned for the subsequent Attachment, responsive, visual-evidence, and final-release Issues.
- Playwright E2E suite: **4 tests passed** (2 E2E-01 tests from Feature 18 + 2 E2E-02 tests from Feature 19).
- `npm run e2e` from `client/`: **passed** (4 tests, Chromium; real Vite client, Express API, PostgreSQL, migrations, seed, and Attachment storage).
- Coverage includes real Requester A/B ownership isolation, active Ticket creation for both contexts, My Tickets search, combined Category/System/Priority/Status filters, Ticket-number sorting and direction, page-size and next-page pagination, owned Ticket Detail rendering, preservation of non-default My Tickets query values after Back, and safe cross-owner/missing Ticket `404` responses without owner data leakage.

Feature 20 Evidence

- Scope is Issue #40 / E2E-03 only; E2E-04 through E2E-06 remain planned for responsive, visual-evidence, and final-release verification.
- Playwright E2E suite: **5 tests passed** (2 E2E-01 tests, 2 E2E-02 tests, and 1 E2E-03 test).
- `npm run e2e` from `client/`: **passed** (5 tests, Chromium; real Vite client, Express API, PostgreSQL, migrations, seed, and isolated Attachment storage).
- Coverage includes valid PDF upload, simulated failed upload with individual Retry, active metadata verification, real byte/MIME/`nosniff` download, valid removal reason, retained removed metadata/timestamp/reason, suppression of removed actions, safe 404 for removed download/re-removal, and unauthorized download/removal rejection.

Feature 21 Evidence

- Scope is Issue #41 / E2E-04 only; E2E-06 remains planned for final-release verification.
- Playwright responsive suite: **3 tests passed** for Desktop 1440×900, Tablet 820×1180, and Mobile 390×844.
- `npm run e2e` from `client/`: **passed** (8 tests total across E2E-01 through E2E-04, Chromium; real Vite client, Express API, PostgreSQL, migrations, seed, and isolated Attachment storage).
- Coverage includes Create Ticket form fit, desktop Ticket table, mobile equivalent Ticket cards, Ticket Detail and Attachment layout, long-filename readability, touch/action visibility, and no page-level horizontal overflow at all required viewports.

Feature 22 Evidence

- Scope is Issue #42 / E2E-05 only; E2E-06 remains planned for final-release verification.
- Visual evidence test: **1 test passed**, producing **29 screenshots** under `artifacts/lab-02/screenshots/` in the required requester-selection, create-ticket, my-tickets, and ticket-detail directories.
- Screenshots cover selector ready/loading/failure; Create Ticket initial/validation/submitting/success/API failure/invalid Attachment; My Tickets A/B/search-filter-sort/page/empty/no-results/failure; owned Detail/upload/download/removal dialog/removed metadata/blocked actions/unauthorized result; and Desktop, Tablet, and Mobile layouts.
- The approved visual checklist in Section 7 is complete: Zen Green tokens, field states, actions, badges, equivalent table/cards, Attachment states, focus, clipping, overlap, filename readability, and page overflow were inspected against `ui-spec.md`.

Feature 23 Evidence

- Scope is Issue #43 / E2E-06 final verification and release preparation.
- Verification baseline: `lab2-staging` commit `6b67a23` (Feature 22 merged); the final documentation commit is recorded in this branch and contains no production-code changes.
- Server: `npm run build` passed; `RUN\_DB\_INTEGRATION=1 npm test` passed (**11 files, 62 tests**).

- Database: `npx prisma migrate deploy` passed with no pending migrations; `npm run prisma:seed` passed twice with no duplicate reference data.
- Client: `npm run build` passed; `npm test -- --run` passed (**9 files, 54 tests**).
- E2E: `npm run e2e` passed (**9 tests**, including E2E-01 through E2E-05 and the 29-screenshot visual evidence test).
- Final audit confirmed no `.env`/secret files, generated build output, Playwright report/test-results, or uploaded storage files are tracked. E2E-06 is now `PASS`; the release PR should use `lab2-staging` as base source and `main` as target.
- Final delivery PDF: `output/pdf/67070507210\_Lab2\_Final.pdf` with `Answer Part 1` through `Answer Part 9` and working GitHub links.

10. Known Limitations and Deferred Tests

- Real authentication/session/token/password/role security belongs to Lab 3; Development Requester tests do not claim real security.
- IT Staff workflows, comments, notes, Actions Taken, and status transitions are excluded.
- Malware scanning and production cloud storage are deferred; Lab 2 still tests type/signature/size/safe naming/non-public storage/ownership/removal.
- Automation does not replace required Human screenshot inspection.

11. Human Approval Checklist

- [x] Test IDs and actual intended file paths are approved.
- [x] Every AC has meaningful planned coverage.
- [x] Failure, boundary, ownership, multi-Requester, responsive, and Attachment cases are sufficient.
- [x] No test result is marked Pass before it runs.
- [x] Human agrees that planned tests prove the approved contract rather than AI-generated assumptions.

Answer Part 4: AI Use with Reflection

LLM used: OpenAI Codex through ChatGPT (GPT 5.6 Sol, High). The AI helped with requirements analysis, repository inspection, contract drafting, test planning, implementation assistance, review diagnosis, evidence checks, and documentation consistency.

Rendered AI record: <docs/lab-02/ai-use.md>

The document contains 7 selected prompts actually used, descriptions of how results were used, mistakes that were rejected, Human approval points, and the student's own reflection.

My Reflection

Lab 2 used an AI agent for repetitive planning and implementation support. I learned to write more focused prompts, constrain each task to one Issue and branch, and reject work that was not explicitly authorized. I retained responsibility for approving requirements, reviewing code and tests, validating evidence, and making the final submission claims.

Rendered copy: docs/lab-02/ai-use.md

[docs/lab-02/ai-use.md](#)

Lab 2 AI Use and Reflection

AI Model and Agent Environment

- **AI product/agent:** OpenAI Codex through ChatGPT
- **Model:** `GPT 5.6 Sol (High)`
- **Primary roles used:** requirements analysis, Lab 1 repository inspection, Git/PR diagnosis, engineering-contract preparation, test planning, UI/API specification, and consistency review

How AI Was Used

AI was used as a specification and engineering assistant. It was not authorized to approve the contract, fabricate peer review, invent evidence, commit, merge, or push the final documents without Human review. The student remains responsible for every requirement, business rule, acceptance criterion, design decision, test, code change, and submission claim.

Selected Key Prompts Actually Used

# Selected prompt (summarized from the actual conversation) How the result was used / Human action required	
--- --- ---	
1	"Lab 2 [REDACTED] STEP [REDACTED] Human [REDACTED] AI [REDACTED]" Used to identify the complete Lab 2 scope, fixed constraints, excluded features, required documents, Git workflow, test evidence, and Human responsibility before implementation. The student must verify the interpretation against the PDF.
2	"Lab [REDACTED] Project [REDACTED] Lab 1 [REDACTED] STEP [REDACTED]" Used to inspect the existing Lab 1 foundation and to enforce the Engineering Contract gate before code. An incorrect workspace path was initially inspected; the student later provided the correct repository path.
3	"[REDACTED] Main [REDACTED] `C:\Users\mos28\Desktop\ [REDACTED]\TokTickIT Full-Stack\Lab1\toktickit` [REDACTED] Step [REDACTED]" Used to verify the real repository, clean `main`, Lab 1 commit history, technology stack, schema, APIs, tests, and documentation. The first required Lab 2 action was identified as the Engineering Contract Issue and branch workflow.
4	"[REDACTED] AI Agent [REDACTED] Human [REDACTED]" Used to create the ordered workflow, Human approval gates, feature dependencies, TDD sequence, PR review rules, integration checks, and final PDF evidence plan.
5	"[REDACTED] PR [REDACTED] [REDACTED] `C:\Users\mos28\Desktop\ [REDACTED]\TokTickIT Full-Stack\Lab1\toktickit` [REDACTED] Path [REDACTED] Pull request #13" Used to inspect PR #13 read-only. The review found correct base/head branches and Issue linkage, but also found six empty Markdown files, a self-referencing `Related to #13`, inaccurate completed checkboxes, and a real Changes Requested review.
6	"[REDACTED] `.md` [REDACTED] AI Agent [REDACTED] Human [REDACTED] PDF [REDACTED]" Used to confirm that AI is expected to help prepare `specification.md`, `tests.md`, `ui-spec.md`, and `api-spec.md`, while Human must review, correct, decide, and approve. It also identified that `reviewer.md` must contain real evidence and that the reflection must be the student's own experience.
7	"[REDACTED] approve [REDACTED] push [REDACTED]" Used to prepare local versions of all six Lab 2 Markdown files on `feature/5-lab2-engineering-contract`. No commit or push occurred during that action; Human approval was required first.

Important AI Corrections and Rejections

1. The AI initially inspected and created files in an incorrect temporary repository path before the student supplied the correct repository path. Those files were not pushed to the student's GitHub repository.
2. The student corrected the working repository to `C:\Users\mos28\Desktop\ [REDACTED]\TokTickIT Full-Stack\Lab1\toktickit`.
3. PR #13 was inspected rather than accepted at face value; the files were confirmed to be empty even though its verification checklist was marked complete.

Human Decision Review

****Reviewed by the student on 2026-08-24:****

- The API contract and Business Rules were reviewed and approved as written.
- Decisions D-01 through D-13 in `specification.md` were approved as written.
- The Test Plan and UI Specification were reviewed and approved as written.

My Reflection

Lab นี้ใช้ AI Agent ทำงานเป็นส่วนใหญ่ เช่นการวางแผนการทำงานที่ซับซ้อน โดยจะเน้นการ prompt ที่ดีกว่า Lab 1 เช่นลงลึกเจาะบทบาทให้มันทำงานเฉพาะส่วนให้ดีมากขึ้นส่วนที่ปฏิเสธมันเด็ดขาดคือการให้มันอย่าทำงานเองโดยมันยังไม่ได้สั่ง

Human Approval Checklist

- [x] The exact AI/model name is confirmed.
- [x] The 6-10 selected prompts represent prompts that were actually used.
- [x] The "How the result was used" descriptions are accurate.
- [x] Mistakes and rejected AI output are not hidden.
- [x] The reflection is written by the student in their own words.
- [x] No AI claim is treated as evidence without repository/test/PR verification.

Answer Part 5: Development Requester Select Screen

This screen simulates the logged-in Requester context for Lab 2 testing only. It loads active Requesters, excludes inactive data, supports loading/failure recovery, and shows the current Requester after Continue.

Requester selector ready with active-user dropdown

TOKTICKIT · LAB 2

Select a Development Requester

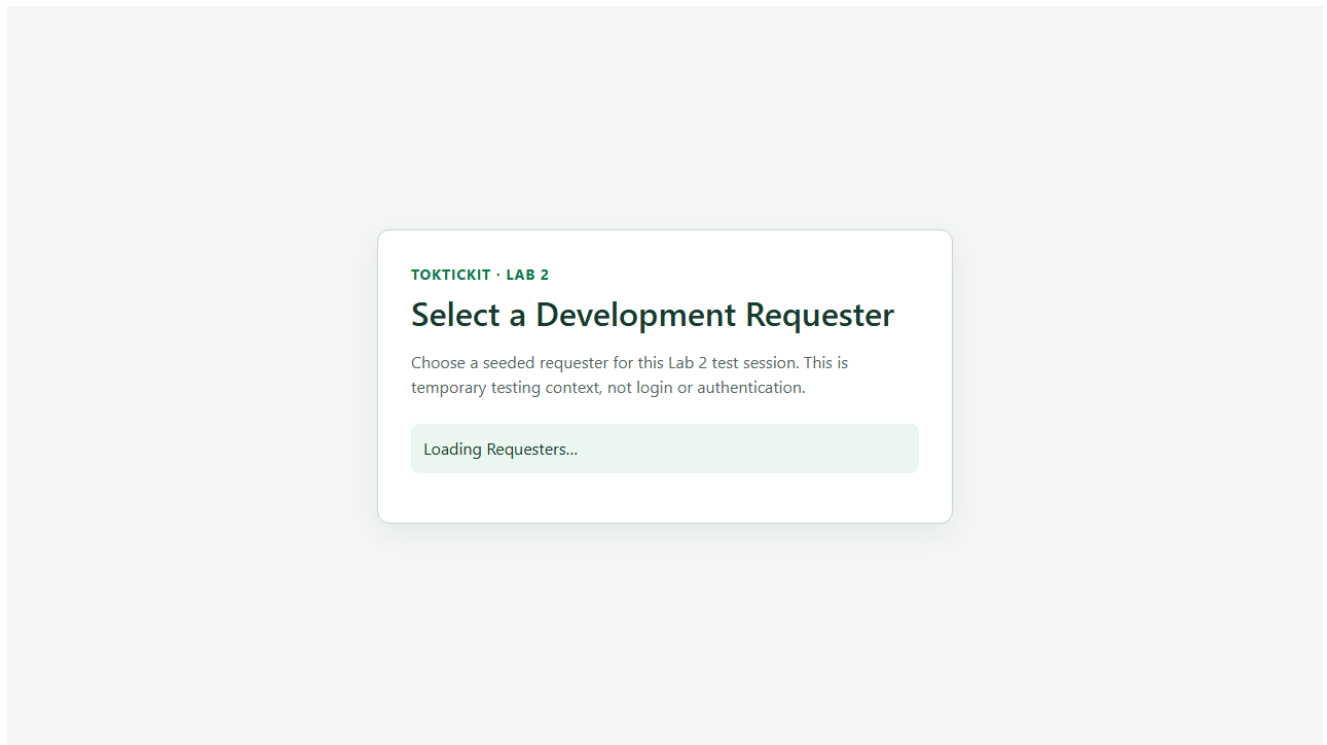
Choose a seeded requester for this Lab 2 test session. This is temporary testing context, not login or authentication.

Development Requester

Select a requester... ▼

Continue

Requester selector loading state



Requester selector safe failure and Retry

TOKTICKIT · LAB 2

Select a Development Requester

Choose a seeded requester for this Lab 2 test session. This is temporary testing context, not login or authentication.

Unable to load Development Requesters. Please try again.

Retry

Answer Part 6: Working Ticket Screen - Create Mode

Create Ticket evidence covers the required initial, validation failure, submitting, success, API failure, and invalid Attachment states. The Requester is selected before entering the application; reference data is loaded from PostgreSQL; Ticket Number, saved values, requesterId, and NEW status come from the backend.

Backend/API proof: [Create Ticket contract](#) | [E2E-01 test](#)

Initial desktop form with Requester and database reference data

TokTickIT [Workspace](#) [My Tickets](#) [Create Ticket](#) Requester: Anan Srisuk [Change Requester](#)

[← Back to workspace](#)
REQUESTER WORKSPACE

Create Ticket

Describe your IT request. Fields marked * are required.

Ticket Number
Generated after submission

Ticket Date
Recorded after submission

Requester
Anan Srisuk

Category *
Select Category ▼

Related System *
Select Related System ▼

Requested Priority
MEDIUM ▼

Summary *

0/120

Description *

0/5000

Attachments

Choose Files No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 files

[Cancel](#) [Submit Ticket](#)

Invalid submission with field-level messages

TokTickIT [Workspace](#) [My Tickets](#) [Create Ticket](#) Requester: Anan Srisuk [Change Requester](#)

[← Back to workspace](#)
REQUESTER WORKSPACE

Create Ticket

Describe your IT request. Fields marked * are required.

Ticket Number
Generated after submission

Ticket Date
Recorded after submission

Requester
Anan Srisuk

Category *
Select Category

Related System *
Select Related System

Requested Priority
MEDIUM

Summary *

Summary must contain 5 to 120 characters.
0/120

Description *

Description must contain 10 to 5,000 characters.
0/5000

Attachments

Choose Files No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 files

[Cancel](#) [Submit Ticket](#)

Submitting state with duplicate activation blocked

TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Anan Srisuk

Change Requester

[← Back to workspace](#)

REQUESTER WORKSPACE

Create Ticket

Describe your IT request. Fields marked * are required.

Ticket Number

Generated after submission

Ticket Date

Recorded after submission

Requester

Anan Srisuk

Category *

Account and Access

Related System *

Campus Wi-Fi

Requested Priority

HIGH

Summary *

Feature 22 API failure 1788621199622

36/120

Description *

Visual evidence for the Lab 2 requester workflow.

49/5000

Attachments

Choose Files

No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 files

• invalid.txt (1 KiB) — Unsupported file type.

Remove

Cancel

Submitting...

TokTickIT Lab 2 - 67070507210

Page 35

Success state with official backend Ticket Number

TokTickIT [Workspace](#) [My Tickets](#) [Create Ticket](#) Requester: Anan Srisuk [Change Requester](#)

Ticket created

Official Ticket Number: **TKT-2026-000336**

Attachment results

- invalid.txt: Unsupported file type.

[Back to workspace](#) [Create another Ticket](#)

Safe API failure with form values retained

TokTickIT

WorkspaceMy TicketsCreate Ticket

Requester: Anan SrisukChange Requester

[← Back to workspace](#)

REQUESTER WORKSPACE

Create Ticket

Describe your IT request. Fields marked * are required.

Unable to create Ticket

Retry

Ticket Number

Generated after submission

Ticket Date

Recorded after submission

Requester

Anan Srisuk

Category *

Account and Access

Related System *

Campus Wi-Fi

Requested Priority

HIGH

Summary *

Feature 22 API failure 1788621199622

36/120

Description *

Visual evidence for the Lab 2 requester workflow.

49/5000

Attachments

Choose Files

No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 files

invalid.txt (1 KiB) — Unsupported file type.

Remove

Cancel

Submit Ticket

Valid and invalid Attachment result shown per file

TokTickIT [Workspace](#) [My Tickets](#) [Create Ticket](#) Requester: Anan Srisuk [Change Requester](#)

[← Back to workspace](#)
REQUESTER WORKSPACE

Create Ticket

Describe your IT request. Fields marked * are required.

Ticket Number
Generated after submission

Ticket Date
Recorded after submission

Requester
Anan Srisuk

Category *

Select Category

Category is required.

Related System *

Select Related System

Related System is required.

Requested Priority
MEDIUM

Summary *

Summary must contain 5 to 120 characters.
0/120

Description *

Description must contain 10 to 5,000 characters.
0/5000

Attachments

Choose Files

No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 files

invalid.txt (1 KiB) — **Unsupported file type.**

Remove

Cancel

Submit Ticket

Answer Part 7: Working My Tickets Screen

Requester A sees only A-owned Tickets. Switching to Requester B removes A data and loads B data. The evidence covers search, filters, sorting, pagination, empty state, no-results state, failure recovery, and cross-requester rejection.

My Tickets contract: [api-spec.md list behavior](#) | [Issue #39](#)

Requester A selected and A-owned Ticket list

TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Anan Srisuk

Change Requester

REQUESTER WORKSPACE

My Tickets

Create Ticket

Tickets created by Anan Srisuk.

Search

Ticket number or summary

Category

All Categories

Related System

All Systems

Requested Priority

All Priorities

Current Status

All Statuses

Sort by

Last updated

Direction

Newest first

Page size

10

Showing 10 of 270 Tickets

Previous

Page 1 of 27

Next

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Page 40

Requester B selected; A data disappears and B data loads

TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Benjamas Kittipong

Change Requester

REQUESTER WORKSPACE

My Tickets

Create Ticket

Tickets created by Benjamas Kittipong.

Search

Ticket number or summary

Category

All Categories

Related System

All Systems

Requested Priority

All Priorities

Current Status

All Statuses

Sort by

Last updated

Direction

Newest first

Page size

10

Showing 10 of 32 Tickets

Previous

Page 1 of 4

Next

TokTickIT Lab 2 - 67070507210

Page 41

Search, filters, sorting, page controls, and page size

TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Anan Srisuk

Change Requester

REQUESTER WORKSPACE

My Tickets

Create Ticket

Tickets created by Anan Srisuk.

Search

al evidence 1788621197907-120884c9

Category

All Categories

Related System

All Systems

Requested Priority

HIGH

Current Status

All Statuses

Sort by

Ticket number

Direction

Oldest first

Page size

10

Clear Filters

Showing 1 of 1 Tickets

Previous

Page 1 of 1

Next

Requester-owned empty state

TokTickIT

Workspace

[My Tickets](#)

Create Ticket

Requester: Anan Srisuk

Change Requester

REQUESTER WORKSPACE

My Tickets

Create Ticket

Tickets created by Anan Srisuk.

Search

Ticket number or summary

Category

All Categories

Related System

All Systems

Requested Priority

All Priorities

Current Status

All Statuses

Sort by

Last updated

Direction

Newest first

Page size

10

You have not created any tickets yet.

Create Ticket

No-results state with Clear Filters

TokTickIT

Workspace

[My Tickets](#)

Create Ticket

Requester: Anan Srisuk

Change Requester

REQUESTER WORKSPACE

My Tickets

Create Ticket

Tickets created by Anan Srisuk.

Search

no-ticket-matches-this-value

Category

All Categories

Related System

All Systems

Requested Priority

HIGH

Current Status

All Statuses

Sort by

Ticket number

Direction

Oldest first

Page size

10

Clear Filters

No tickets match the current search or filters.

Clear Filters

TokTickIT Lab 2 - 67070507210

Page 44

Safe Ticket-list failure with Retry

TokTickIT

Workspace

[My Tickets](#)

Create Ticket

Requester: Anan Srisuk

Change Requester

REQUESTER WORKSPACE

My Tickets

Create Ticket

Tickets created by Anan Srisuk.

Search

Ticket number or summary

Category

All Categories

Related System

All Systems

Requested Priority

All Priorities

Current Status

All Statuses

Sort by

Last updated

Direction

Newest first

Page size

10

Unable to load Tickets.

Retry

Cross-requester direct-access evidence returns safe not-found

TokTickIT [Workspace](#) [My Tickets](#) [Create Ticket](#) Requester: Benjamas Kittipong [Change Requester](#)

[← Back to My Tickets](#)

Ticket not found or unavailable.

[Retry](#)

Answer Part 8: Working Ticket Screen - View Mode and Attachments

Owned Ticket Detail is read-only and shows Ticket information plus Attachment metadata. Evidence covers adding an Attachment, active download, individual retry, soft removal with a valid reason, retained removed metadata, blocked removed download, and unauthorized access rejection.

Attachment and ownership contract: [api-spec.md Attachment endpoints](#) | [Issue #40](#)

Owned Ticket Detail with active Download and Remove

TokTickIT

Workspace

[My Tickets](#)

[Create Ticket](#)

Requester: Anan Srisuk

Change Requester

[← Back to My Tickets](#)

REQUESTER WORKSPACE

TKT-2026-000335

NEW

Ticket Detail for Anan Srisuk

Ticket Information

Ticket Number

TKT-2026-000335

Requester

Anan Srisuk (anan.srisuk@example.test)

Related System

Campus Wi-Fi

Current Status

NEW

Summary

Feature 22 visual evidence 1788621197907-120884c9

Description

Visual evidence for the Lab 2 requester workflow.

Ticket Date

Sep 5, 2026, 10:13 PM

Category

Account and Access

Requested Priority

HIGH

Last Updated

Sep 5, 2026, 10:13 PM

Attachments

Add Attachment

Choose Files

No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 active files

visual-evidence-1788621197907-120884c9.pdf

application/pdf · 1 KiB · Uploaded Sep 5, 2026, 10:13 PM

Active

Download

Remove

Add Attachment and pending upload state

TokTickIT

Workspace

[My Tickets](#)

[Create Ticket](#)

Requester: Anan Srisuk

Change Requester

[← Back to My Tickets](#)

REQUESTER WORKSPACE

TKT-2026-000335

NEW

Ticket Detail for Anan Srisuk

Ticket Information

Ticket Number

TKT-2026-000335

Requester

Anan Srisuk (anan.srisuk@example.test)

Related System

Campus Wi-Fi

Current Status

NEW

Summary

Feature 22 visual evidence 1788621197907-120884c9

Description

Visual evidence for the Lab 2 requester workflow.

Ticket Date

Sep 5, 2026, 10:13 PM

Category

Account and Access

Requested Priority

HIGH

Last Updated

Sep 5, 2026, 10:13 PM

Attachments

Add Attachment

Choose Files

No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 active files

pending-visual.pdf

1 KiB

Upload

Remove

visual-evidence-1788621197907-120884c9.pdf

application/pdf · 1 KiB

Uploaded Sep 5, 2026, 10:13 PM

Active

Download

Remove

Soft-removal dialog with required reason and focus

TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Anan Srisuk

Change Requester

← Back to My Tickets

REQUESTER WORKSPACE

TKT-2026-000335

NEW

Ticket Detail for Anan Srisuk

Ticket Information

Ticket Number

TKT-2026-000335

Requester

Anan Srisuk (anan.srisuk@example.test)

Related System

Campus Wi-Fi

Ticket Date

Sep 5, 2026, 10:13 PM

Category

Account and Access

Requested Priority

HIGH

Current Status

NEW

Last Updated

Sep 5, 2026, 10:13 PM

Summary

Feature 22 visual evidence 1788621197907-120884c9

Description

Visual evidence for the Lab 2 requester workflow

Attachments

Add Attachment

Choose Files

No file chosen

JPG, JPEG, PNG, WEBP, or PDF;

pending-visual.pdf

1 KiB

visual-evidence-1788621197907-120884c9.pdf

application/pdf · 1 KiB

Uploaded Sep 5, 2026, 10:13 PM

Active

Download

Remove

Remove visual-evidence-1788621197907-120884c9.pdf?

This will hide the file while retaining its metadata.

Reason

Cancel

Confirm removal

Upload

Remove

TokTickIT Lab 2 - 67070507210

Page 50

Retained removed metadata; Download/Remove actions blocked

TokTickIT

Workspace

[My Tickets](#)

Create Ticket

Requester: Anan Srisuk

Change Requester

[← Back to My Tickets](#)

REQUESTER WORKSPACE

TKT-2026-000335

NEW

Ticket Detail for Anan Srisuk

Ticket Information

Ticket Number
TKT-2026-000335

Requester
Anan Srisuk (anan.srisuk@example.test)

Related System
Campus Wi-Fi

Current Status
NEW

Summary
Feature 22 visual evidence 1788621197907-120884c9

Description
Visual evidence for the Lab 2 requester workflow.

Ticket Date
Sep 5, 2026, 10:13 PM

Category
Account and Access

Requested Priority
HIGH

Last Updated
Sep 5, 2026, 10:13 PM

Attachments

Add Attachment

Choose Files

No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 active files

visual-evidence-1788621197907-120884c9.pdf

application/pdf · 1 KiB

Uploaded Sep 5, 2026, 10:13 PM

Removed · Visual evidence cleanup

Removed at Sep 5, 2026, 10:13 PM

Unauthorized Ticket or Attachment access rejected safely

TokTickIT

Workspace

[My Tickets](#)

[Create Ticket](#)

Requester: Benjamas Kittipong

[Change Requester](#)

[← Back to My Tickets](#)

Ticket not found or unavailable.

[Retry](#)

Answer Part 9: Zen Green UI and Responsive Evidence

The UI follows the approved Zen Green palette, readable text badges, editable/read-only distinctions, validation placement, button hierarchy, and responsive rules. The visual checklist was completed against ui-spec.md.

Rendered UI specification: <docs/lab-02/ui-spec.md>

Checklist verified: colors; editable/read-only fields; required markers; validation messages; primary/secondary/busy/destructive buttons; badges; equivalent table/cards; clipping; overlap; readable filenames; hidden actions; and page-level horizontal overflow.

Rendered copy: docs/lab-02/ui-spec.md

[docs/lab-02/ui-spec.md](#)

Lab 2 Zen Green UI Specification

1. Design Principles

- Professional, calm, readable, and consistent with TokTickIT's Zen Green direction.
- Current Requester identity and available actions are always clear.
- Editable and read-only values remain distinguishable without relying only on color.
- Validation stays close to the affected field and recovery actions are explicit.
- Desktop/mobile representations may differ but provide equivalent information/actions.
- Reusable components are preferred over screen-specific styling.

2. Design Tokens

Colors

Token	Value	Use
---	---	---
<code>--color-green-primary</code>	<code>#006B3C</code>	Header, primary buttons, strong emphasis
<code>--color-green-secondary</code>	<code>#0B7A46</code>	Active navigation, focus, links, hover
<code>--color-green-pale</code>	<code>#EAF6EF</code>	Selected/success/subtle emphasis
<code>--color-page</code>	<code>#F5F7F6</code>	Page background
<code>--color-surface</code>	<code>FFFFFF</code>	Cards, panels, editable controls
<code>--color-text</code>	<code>#173B2C</code>	Primary charcoal-green text
<code>--color-text-muted</code>	<code>#52685E</code>	Secondary text
<code>--color-border</code>	<code>#C9D6CF</code>	Neutral boundaries
<code>--color-readonly</code>	<code>#EFF3F0</code>	Read-only fields
<code>--color-error</code>	<code>#9B1C1C</code>	Error text/border
<code>--color-warning</code>	<code>#9A6700</code>	Warning only
<code>--color-warning-bg</code>	<code>FFF4CE</code>	Warning background
<code>--color-focus</code>	<code>#0B7A46</code>	Keyboard focus ring

Success, warning, priority, status, error, active, and removed states always include text or accessible names; color alone is insufficient.

Typography and Spacing

- Font stack: `Inter`, `Segoe UI`, `Roboto`, system sans-serif.
- Body: 16 px, line-height 1.5; small/meta text minimum 14 px.
- Page title: 28-32 px, weight 700; section title: 20-24 px, weight 650-700.
- Form labels: 14-16 px, weight 600; placeholders never replace labels.
- Spacing scale: 4, 8, 12, 16, 24, 32, 48 px.
- Default control height and touch target: at least 44 px where practical.
- Card radius: 8-12 px; button/control radius: 6-8 px.
- Surfaces use white, subtle border, and restrained shadow.

3. Application Shell and Navigation

Desktop and Tablet

- Primary Green header with TokTickIT identity.
- Navigation contains My Tickets and Create Ticket.
- Active page is visually clear and uses `aria-current="page"`.
- Current Development Requester name and Change Requester action remain visible.
- Header content aligns with the page maximum width.

Mobile

- TokTickIT identity remains visible.
- Navigation collapses into an accessible menu or compact stacked layout.
- Controls have text/accessible names and visible focus.
- Current Requester and Change Requester remain reachable without horizontal scrolling.

Route Guard

- No valid Requester: render the selector without flashing requester-owned content.
- While restoring/validating selection: show a labelled page-level loading state.

4. Development Requester Selection

Required content:

- TokTickIT title.
- `Select a Development Requester` heading.
- Explanation that this is Lab 2 testing only, not login/authentication.
- Labelled dropdown populated from PostgreSQL.
- Continue button disabled until a Requester is selected.

States:

- **Loading:** disabled controls, `Loading Requesters...`, `aria-busy`.
- **Ready:** active Requesters in deterministic order.
- **Empty:** no-active-Requester explanation and Retry.
- **Failure:** safe inline alert and Retry.
- **Validating:** Continue uses busy text and blocks duplicate activation.

Use a centered surface with maximum width around 560 px and natural mobile stacking.

5. Shared Form Components

Labels and Required Markers

- Labels appear above controls with consistent spacing.
- Required fields show a red asterisk and semantic required state.
- A nearby legend explains `\*`; the asterisk never replaces validation text.

Editable, Read-only, Invalid, and Disabled Controls

- Editable: white background, neutral border, dark text.
- Read-only: soft gray-green background, readable text, and hint such as `Generated after submission` when needed.
- Focus: visible 2-3 px green ring with sufficient contrast.
- Invalid: dark red border/text; message immediately below and linked using `aria-describedby`.
- Disabled: visibly muted, semantically disabled, and non-activatable.
- Invalid submit moves focus to the first invalid field.

Buttons

- Primary: solid Primary Green; one dominant action per section.
- Secondary: white/pale surface with green border/text.
- Tertiary: text/link style for low-emphasis actions.
- Destructive: dark red styling only for confirmed removal.
- Busy: retains size, shows `Submitting...`, `Uploading...`, or `Removing...`, exposes busy semantics, and disables repeat activation.

- Icon-only controls require an accessible label and tooltip.

6. Create Ticket Screen

Information order:

1. Heading and instruction.
2. Read-only Ticket Number, Ticket Date, and Requester.
3. Category, Related System, and Requested Priority.
4. Summary and Description.
5. Attachment selection and per-file results.
6. Safe form/partial-failure feedback.
7. Secondary Cancel/Back and primary Submit Ticket actions.

Field behavior:

- Ticket Number: `Generated after submission` before success.
- Ticket Date: `Recorded after submission` before success.
- Requester: current selected Requester; never editable on this form.
- Requested Priority: default Medium, still editable.
- Summary receives sufficient/full width when constrained.
- Description is multi-line; vertical resize only if layout remains intact.
- Accessible character counts appear near approved limits.

Attachment selection:

- Hint: `JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 active files`.
- Each selected file shows filename, formatted size, state, and Remove-from-selection.
- Invalid files remain visible with their reason and are never submitted.
- Exceeding five shows an actionable error.
- Long filenames wrap/truncate visually without page overflow; full name remains accessible.

Screen states:

- **Initial/loading:** dependent controls disabled until reference data is ready.
- **Validation failure:** field messages, first invalid focus, values retained.
- **Submitting:** mutating controls disabled and button reads `Submitting...`.
- **Success:** confirmation with official backend Ticket Number and `View Ticket`/`Go to My Tickets`.
- **API failure:** safe alert, values/files retained, Retry available.
- **Partial Attachment failure:** Ticket-success confirmation plus per-file results and Ticket Detail/retry action.

7. My Tickets

Toolbar

- Title, selected Requester context, and primary Create Ticket action.
- Labelled search input.
- Category, Related System, Requested Priority, and Current Status filters.
- Sort field/direction.
- Clear Filters available only when non-default criteria exist.
- Search/filter change resets to page 1.

Desktop Table

Columns:

- Ticket Number
- Summary
- Category
- Related System
- Requested Priority
- Current Status
- Last Updated
- Explicit View action

Use semantic headers and expose sort direction through text/ARIA.

Mobile Cards

Each card includes Ticket Number, Summary, Category, Related System, Requested Priority, Current Status, Last Updated, and a touch-friendly View Ticket action.

Pagination

- Show current page, total pages, and total item count.
- Previous/Next and page controls have accessible names and disabled boundary states.
- Page sizes: 10, 20, 50; changing size resets to page 1.
- No horizontal overflow.

Result States

- Loading: labelled state; old Requester data is not shown as new data.
- Empty: 'You have not created any tickets yet' plus Create Ticket.
- No results: 'No tickets match the current search or filters' plus Clear Filters.
- Failure: safe alert, Retry, and preserved controls.
- Invalid query: recoverable message and Reset to defaults.

8. Requester Ticket Detail

- Back to My Tickets while preserving appropriate list state within the same Requester context.
- Ticket Number as the primary identifier.
- Ticket Date, Requester, Category, Related System, Summary, Requested Priority, Description, and Current Status are read-only.
- Use labelled cards/definition lists where practical rather than editable-looking controls.
- Do not render Public Comments, Internal Notes, Actions Taken, Event Log, IT Staff controls, status controls, or resolution workflow.

Badge rules:

- Every badge contains text.
- Requested Priority: Low, Medium, High, Urgent.
- Current Status in Lab 2: New.
- IT Priority may have a future component variant, but no editable Lab 2 workflow.

9. Attachment Section

Active

- Filename, type, size, uploaded timestamp.
- Download and owner Remove actions.
- Optional safe browser preview may be considered only after Human approval; download remains required.

Uploading/Invalid/Unavailable

- Per-file busy/result state and duplicate prevention.

- Success replaces pending data with authoritative API metadata.
- Failure retains filename and shows Retry/Remove-from-queue.
- Storage failure displays `File temporarily unavailable` without server paths.

Removed

- Metadata remains with textual `Removed` badge, removed time, and reason.
- No Download, Preview, or Remove-again action.

Removal Confirmation

- Accessible dialog names the file and explains the result.
- Reason required, 5-250 characters.
- Focus enters/traps in dialog and returns to trigger on close.
- Confirm uses destructive busy state `Removing...`.

10. Feedback Components

- Inline field errors are associated with one control.
- Page alerts provide safe failure and Retry when meaningful.
- Success remains long enough to read and includes next action.
- Warning callouts are reserved for partial upload failure or unsaved-change confirmation.
- Empty/no-results states include explanation and useful action.
- Live regions announce important asynchronous updates without repeated noise.

11. Responsive Layout

Viewport	Required behavior
---	---
Desktop `>= 992px`	Centered max-width around 1200 px; multi-column form; full Ticket table
Tablet `768-991px`	Two columns where practical; Summary, Description, Attachments full width when needed
Mobile `<< 768px`	Stacked fields/actions; My Tickets cards; touch-friendly controls; no page-level horizontal scroll

At all sizes:

- No clipped labels/messages, overlap, hidden actions, or unreadable filenames.
- Dialog actions remain reachable.
- Browser zoom to 200% preserves required content/function.

12. Accessibility

- Logical semantic headings and one primary page heading.
- Programmatic labels, descriptions, required state, and invalid state.
- Complete keyboard operation and visible focus.
- Active navigation uses semantics in addition to color.
- Table headers/sort state are exposed.
- Loading/busy/result updates are announced appropriately.
- Accessible dialog focus management.
- WCAG 2.1 AA contrast for normal text and meaningful controls.
- Reduced-motion preference respected.
- No state relies only on color.

13. Screenshot Paths

```
artifacts/lab-02/screenshots/
■■■ requester-selection/
■■■ create-ticket/
```

■■■ my-tickets/
■■■ ticket-detail/

Evidence must include:

- Selector ready, loading, and failure.
- Create initial, validation, submitting, success, API failure, and invalid Attachment.
- My Tickets A/B, search/filter/sort/page, empty, no-results, and failure.
- Detail owned view, upload/download, removal dialog, removed metadata, blocked download, and unauthorized result.
- Desktop, tablet, and mobile captures for Create Ticket, My Tickets, and Ticket Detail.

14. Visual Inspection Checklist

- [] Color tokens match approved Zen Green rules.
- [] Editable/read-only fields are distinct.
- [] Required markers and validation messages are consistent.
- [] Button hierarchy and busy/destructive/disabled states are clear.
- [] Active page and current Requester are understandable.
- [] Priority/status badges include readable text.
- [] Empty and no-results states differ.
- [] Desktop table and mobile cards contain equivalent essential information.
- [] Attachment states are understandable.
- [] No clipping, overlap, hidden control, unreadable filename, or page overflow.
- [] Keyboard focus is visible and logical.
- [] Human compared screenshots with this approved specification.

15. Human Approval Items

Approval status (2026-08-24): Approved by the student.

Create Ticket - Desktop 1440x900

TokTickIT [Workspace](#) [My Tickets](#) [Create Ticket](#) Requester: Anan Srisuk [Change Requester](#)

[← Back to workspace](#)
REQUESTER WORKSPACE

Create Ticket

Describe your IT request. Fields marked * are required.

Ticket Number

Generated after submission

Ticket Date

Recorded after submission

Requester

Anan Srisuk

Category *

Select Category ▼

Related System *

Select Related System ▼

Requested Priority

MEDIUM ▼

Summary *

0/120

Description *

0/5000

Attachments

Choose Files

 No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 files

Cancel

Submit Ticket

Create Ticket - Tablet 820x1180

TokTickIT [Workspace](#) [My Tickets](#) [Create Ticket](#) Requester: Anan Srisuk [Change Requester](#)

[← Back to workspace](#)

REQUESTER WORKSPACE

Create Ticket

Describe your IT request. Fields marked * are required.

Ticket Number

Generated after submission

Ticket Date

Recorded after submission

Requester

Anan Srisuk

Category *

Select Category ▼

Related System *

Select Related System ▼

Requested Priority

MEDIUM ▼

Summary *

0/120

Description *

0/5000

Attachments

Choose Files

No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 files

Cancel

Submit Ticket

Create Ticket - Mobile 390x844

TokTickIT

WorkspaceMy TicketsCreate Ticket

Requester: Anan SrisukChange Requester

[← Back to workspace](#)**REQUESTER WORKSPACE**

Create Ticket

Describe your IT request. Fields marked * are required.

Ticket Number

Generated after submission

Ticket Date

Recorded after submission

Requester

Anan Srisuk

Category *

Select Category

Related System *

Select Related System

Requested Priority

MEDIUM

Summary *

0/120

Description *

0/5000

Attachments

Choose Files

No file chosen

JPG, JPEG, PNG, WEBP, or PDF; maximum 5 MiB each; maximum 5 files

Submit Ticket

Cancel

TokTickIT Lab 2 - 67070507210

Page 62

My Tickets - Desktop table

TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Anan Srisuk

Change Requester

REQUESTER WORKSPACE

My Tickets

Create Ticket

Tickets created by Anan Srisuk.

Search

Feature 22 visual evidence 178862120:

Category

All Categories

Related System

All Systems

Requested Priority

All Priorities

Current Status

All Statuses

Sort by

Last updated

Direction

Newest first

Page size

10

Clear Filters

Showing 1 of 1 Tickets

Previous

Page 1 of 1

Next

My Tickets - Tablet layout

TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Anan Srisuk

Change Requester

REQUESTER WORKSPACE

My Tickets

Create Ticket

Tickets created by Anan Srisuk.

Search

Feature 22 visual evidence 1788621204285-f34f331e

Category

All Categories

Related System

All Systems

Requested Priority

All Priorities

Current Status

All Statuses

Sort by

Last updated

Direction

Newest first

Page size

10

Clear Filters

Showing 1 of 1 Tickets

Ticket Number	Summary	Category	Related System	Requested Priority	Current Status	Last Updated
TKT-2026-000338	Feature 22 visual evidence 1788621204285-f34f331e	Account and Access	Campus Wi-Fi	HIGH	NEW	Sep 5, 2026, 10 PM

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Page 1 of 1

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My Tickets - Mobile equivalent cards

TokTickIT

[Workspace](#) [My Tickets](#) [Create Ticket](#)

Requester: Anan Srisuk [Change Requester](#)

REQUESTER WORKSPACE

My Tickets

Tickets created by Anan Srisuk.

Create Ticket

Search

Feature 22 visual evidence 1788621205147-21b

Category

All Categories

Related System

All Systems

Requested Priority

All Priorities

Current Status

All Statuses

Sort by

Last updated

Direction

Newest first

Page size

10

Clear Filters

Showing 1 of 1 Tickets

TKT-2026-000339

Summary

Feature 22 visual evidence 1788621205147-21b9aee2

Category

Account and Access

Related System

Campus Wi-Fi

Requested Priority

HIGH

Current Status

NEW

Last Updated

Sep 5, 2026, 10:13 PM

View Ticket

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Ticket Detail - Desktop

TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Anan Srisuk

Change Requester

← Back to My Tickets

REQUESTER WORKSPACE

TKT-2026-000337

Ticket Detail for Anan Srisuk

NEW

Ticket Information

Ticket Number

TKT-2026-000337

Requester

Anan Srisuk (anan.srisuk@example.test)

Related System

Campus Wi-Fi

Current Status

NEW

Summary

Feature 22 visual evidence 1788621203371-844779ae

Description

Visual evidence for the Lab 2 requester workflow.

Ticket Date

Sep 5, 2026, 10:13 PM

Category

Account and Access

Requested Priority

HIGH

Last Updated

Sep 5, 2026, 10:13 PM

Attachments

Add Attachment

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visual-evidence-1788621203371-844779ae.pdf

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TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Anan Srisuk

Change Requester

← Back to My Tickets

REQUESTER WORKSPACE

TKT-2026-000338

NEW

Ticket Detail for Anan Srisuk

Ticket Information

Ticket Number

TKT-2026-000338

Ticket Date

Sep 5, 2026, 10:13 PM

Requester

Anan Srisuk (anan.srisuk@example.test)

Category

Account and Access

Related System

Campus Wi-Fi

Requested Priority

HIGH

Current Status

NEW

Last Updated

Sep 5, 2026, 10:13 PM

Summary

Feature 22 visual evidence 1788621204285-f34f331e

Description

Visual evidence for the Lab 2 requester workflow.

Attachments

Add Attachment

Choose Files

No file chosen

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Ticket Detail - Mobile with readable Attachment filename and actions

TokTickIT

Workspace

My Tickets

Create Ticket

Requester: Anan Srisuk

Change Requester

← Back to My Tickets

REQUESTER WORKSPACE

TKT-2026-000339

Ticket Detail for Anan Srisuk

NEW

Ticket Information

Ticket Number

TKT-2026-000339

Ticket Date

Sep 5, 2026, 10:13 PM

Requester

Anan Srisuk (anan.srisuk@example.test)

Category

Account and Access

Related System

Campus Wi-Fi

Requested Priority

HIGH

Current Status

NEW

Last Updated

Sep 5, 2026, 10:13 PM

Summary

Feature 22 visual evidence 1788621205147-21b9aee2

Description

Visual evidence for the Lab 2 requester workflow.

Attachments

Add Attachment

Choose Files

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